

Certified Alpha Capacity:

STATISTICAL ARBITRAGE WHEN LEARNING TAKES TIME

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Abstract

Statistical validation takes time, and alpha can decay before the evidence justifies deployment. We quantify the surviving opportunity through Certified Alpha Capacity: the maximum expected value remaining under prescribed false-deployment and power constraints. In a canonical Gaussian experiment, we derive an exact certification frontier and the sharp law $\mathcal{K}_{\alpha,\beta}/\log(1/\varepsilon)$ for residual information capacity just above it. Near the frontier, a single interim decision can preserve orders of magnitude more capacity than fixed-time testing. Mapping information into cash reveals why equally certifiable signals can retain different economic values. In the specified competitive equilibrium, lifetime information approaches the certification frontier exponentially as research costs vanish.

AMS 2020 Subject Classification: Primary 62L10, 62L15. Secondary 60G40, 62M10.

Keywords: Certified Alpha Capacity, alpha decay, sequential testing, information theory, quantitative finance, optimal stopping, multiple testing.

1. Introduction

A promising trading signal creates a timing problem before it creates a portfolio position. Evidence takes time to accumulate, and the opportunity can weaken during that time. A large current signal may disappear before it supports a reliable deployment decision; a smaller, persistent signal may permit enough observation to justify action. Statistical arbitrage therefore requires assessing both the evidence needed to act and the opportunity remaining when that evidence arrives.

Research on post-publication attenuation, slow exploitation, and alpha dynamics documents the importance of persistence for investment opportunities [27–29, 35]. Selecting among many candidate signals also increases the burden of establishing a reliable discovery [12–14]. Together, these mechanisms make the value available after validation depend on the relative rates of learning and economic decay.

We study this dependence through *Certified Alpha Capacity* (CAC). A certification contract fixes a maximum false-deployment probability under an economic boundary model and a required deployment probability under the alternative. Kullback–Leibler information measures separation from that boundary and provides a clock for accumulated evidence and remaining information lifetime. A stopping rule chooses when to certify, subject to the contract, and receives the value left at that decision. The framework joins a reliability requirement to the reward from timely action within a specified observation experiment.

Sequential testing and group-sequential design supply the statistical foundation. Likelihood processes control errors as observations arrive [1, 2]; error spending and optimal designs allocate the reliability budget across monitoring dates [3–6]. Opportunistic detection and time-sensitive testing make terminal constraints central to the decision [7–9]. Our central result quantifies the value reliable stopping can preserve when the lifetime is only just sufficient for certification: a sharp inverse-logarithmic law with matching bounds over admissible stopping rules (Theorem 3.2). A small surplus of lifetime information has a quantitative value given by this explicit stopping-value law.

The implications connect statistical feasibility to arbitrage economics. At false-deployment level 0.05 and power 0.90, the sharp constant is $\mathcal{K}_{0.05,0.10} \simeq 11.15$. In the zero-hurdle exponential benchmark, a Bonferroni search over 100 candidates with annualized instantaneous Sharpe 2 requires a half-life of about 7.24 years (Section 5). An ordered family of unknown decay profiles retains an exact least-profile frontier (Theorem 7.1), while competitive entry drives excess lifetime information exponentially toward that frontier as research costs vanish (Corollary 8.2). A descriptive funding benchmark illustrates the link between estimated lifetime information and persistence (Section 9).

Economic interpretation requires a payoff map. Information in a Gaussian drift experiment is quadratic in the drift gap, whereas expected cashflow from a bounded position is linear in that gap. Equal lifetime information can thus support different cash values after learning. This connects CAC to learning with deadlines and decision-valued experimental design [17, 18, 20, 33, 34]. Information acquisition and arbitrage can also change other participants' opportunities [21, 23–25], and statistical limits of arbitrage connect econometric discrimination to market opportunities [22]. CAC isolates the timing constraint within a declared observation law, payoff map, and lifetime mechanism.

2. Model

2.1. Probabilistic setup and information time

On a common observation space, let $\mathbb{F} = (\mathcal{F}_t)$ be the usual right-continuous augmentation of $\sigma\{(X_u, Y_u) : u \leq t\}$, and let $\mathbb{P}_0, \mathbb{P}_1$ be locally equivalent laws. The signal X and loading θ are predictable, and

$$dY_t = \theta_t X_t dt + \sigma dB_t^{(1)} \quad (\mathbb{P}_1), \quad \sigma > 0, \quad (2.1)$$

$$dY_t = \sigma dB_t^{(0)} \quad (\mathbb{P}_0), \quad (2.2)$$

where each $B^{(i)}$ is Brownian in the stated law and filtration. We impose local square integrability and the exponential-martingale conditions for Girsanov (Novikov suffices on bounded horizons). The joint law of (X, Y) is assumed to have the likelihood below: the initial observation has the same law and the predictor supplies no additional likelihood factor. Predictability alone does not imply this joint-law assumption. A common completion is used on each horizon.

Define the continuous information clock and its stopped inverse by

$$A_t = \sigma^{-2} \int_0^t \theta_u^2 X_u^2 du, \quad H = A_T, \quad (2.3)$$

$$t_{\text{inv}}(s) = \inf\{t : A_t > s\} \wedge T, \quad \mathcal{G}_s = \mathcal{F}_{t_{\text{inv}}(s)}. \quad (2.4)$$

Here T is economic death, deterministic or an $\mathbb{F}$ -stopping time; A denotes the value of H when deterministic. The information filtration $\mathbb{G} = (\mathcal{G}_s)$ is constant after H , and flat calendar-clock

intervals are collapsed. Girsanov gives the log likelihood

$$\Lambda_t = \int_0^t \frac{\theta_u X_u}{\sigma} dB_u^{(0)} - A_t/2 \quad (\mathbb{P}_0), \quad (2.5)$$

$$\Lambda_t = \int_0^t \frac{\theta_u X_u}{\sigma} dB_u^{(1)} + A_t/2 \quad (\mathbb{P}_1). \quad (2.6)$$

Thus $Z_s = \Lambda_{t_{\text{inv}}(s)}$ satisfies, up to H ,

$$Z_s = \tilde{B}_s^{(1)} + s/2 \quad (\mathbb{P}_1), \quad Z_s = \tilde{B}_s^{(0)} - s/2 \quad (\mathbb{P}_0), \quad (2.7)$$

$$\frac{d\mathbb{P}_1}{d\mathbb{P}_0} \Big|_{\mathcal{G}_s} = e^{Z_s}. \quad (2.8)$$

At a random horizon the precise identity is stopped: $Z_s = \tilde{B}_{s \wedge H}^{(i)} \pm (s \wedge H)/2$. Any Brownian extension after H is auxiliary on an enlarged space. For a bounded $\mathbb{G}$ -stopping time S , write $\mathbb{P}_i^S$ for the restriction to its stopped sigma-field $\mathcal{G}_S$, and $\mathcal{T}_A$ for the stopping times $0 \leq S \leq A$.

We use $D_{\text{KL}}(P\|Q) = \mathbb{E}_P \log(dP/dQ)$ and $\text{kl}(p, q) = p \log(p/q) + (1-p) \log((1-p)/(1-q))$, extended to boundary arguments by lower semicontinuity. All logarithms are natural; Φ is the standard normal cdf, $\bar{\Phi} = 1 - \Phi$, $z_u = \Phi^{-1}(u)$, and $x_+ = \max(x, 0)$. Matrix inequalities use the positive-definite order. Principal model-specific notation is collected in Appendix G.

2.2. Economic value on the information clock

At a fixed calendar time, suppress the time subscript and write X and θ for the current signal value and drift loading. For the quadratic instantaneous trading objective

$$q\theta X - \frac{\gamma}{2}q^2, \quad \gamma > 0,$$

where $q \in \mathbb{R}$ is the position and γ is the quadratic trading-cost coefficient, the full-information optimal position is $q^* = \theta X/\gamma$. Its value rate is

$$\frac{\theta^2 X^2}{2\gamma} dt = \frac{\sigma^2}{2\gamma} dA_t.$$

Thus statistical information and perfect-information economic value are carried by the same clock. When $H = A$ is deterministic, define the lifetime path information and integrated perfect-information value by

$$\mathcal{I}_{\text{life}} := D_{\text{KL}}(\mathbb{P}_1^A \| \mathbb{P}_0^A), \quad V^{PI} := \int_0^T \frac{\theta_t^2 X_t^2}{2\gamma} dt,$$

where the superscript PI stands for perfect information.

Theorem 2.1 (Statistical information and full-information value). *In the deterministic information-horizon benchmark $H = A$,*

$$\mathcal{I}_{\text{life}} = \frac{A}{2}, \quad V^{PI} = \frac{\sigma^2}{2\gamma} A = \frac{\sigma^2}{\gamma} \mathcal{I}_{\text{life}}.$$

More generally, for any bounded $\mathbb{G}$ -stopping time S satisfying $S \leq H$ almost surely,

$$D_{\text{KL}}(\mathbb{P}_1^S \| \mathbb{P}_0^S) = \mathbb{E}_1[Z_S] = \frac{1}{2} \mathbb{E}_1[S].$$

Proof. For every bounded stopping time S , the stopped likelihood-ratio identity gives

$$\frac{d\mathbb{P}_1}{d\mathbb{P}_0} \Big|_{\mathcal{G}_S} = e^{Z_S}.$$

Therefore

$$D_{\text{KL}}(\mathbb{P}_1^S \| \mathbb{P}_0^S) = \mathbb{E}_1[Z_S].$$

Under $\mathbb{P}_1$, $Z_s = \tilde{B}_s^{(1)} + s/2$. Boundedness of S makes optional sampling applicable to $\tilde{B}^{(1)}$, so $\mathbb{E}_1[\tilde{B}_S^{(1)}] = 0$ and hence $\mathbb{E}_1[Z_S] = \mathbb{E}_1[S]/2$. For deterministic $S = A$, this gives $\mathcal{I}_{\text{life}} = A/2$. The economic identity follows from the full-information quadratic value calculation immediately above. $\square$

Under this reduction, signal amplitude, feature units, calendar half-life and lookback enter the certification problem through the total information generated before the economic endpoint.

3. Reliable certification before decay

Fix error parameters $\alpha, \beta \in (0, 1)$, where α is the maximum false-deployment probability under $\mathbb{P}_0$ and $1 - \beta$ is the required deployment probability under $\mathbb{P}_1$. Unless stated otherwise we consider the nondegenerate regime $1 - \beta \geq \alpha$.

With only a type-I constraint, a procedure can deploy immediately with probability α without learning anything about the signal. A power target strictly above α rules out this lottery and makes certification depend on evidence accumulated under the alternative.

Throughout the canonical experiment, the observation filtration is augmented at time zero by a variable $U_{\text{rnd}} \sim \text{Unif}(0, 1)$, independent of the observed path under both hypotheses and with the same law under both. This permits binary randomized decisions without adding information about the hypothesis.

For deterministic information lifetime A , a certification/deployment rule is a pair (S, δ) , where $S \in \mathcal{T}_A$ and $\delta \in \{0, 1\}$ is $\mathcal{G}_S$ -measurable. The event $\{\delta = 1\}$ means that the signal is certified and deployed. On $\{\delta = 0\}$ we may, without loss, set $S = A$. Define

$$c_{\alpha, \beta}(A) := \sup \mathbb{E}_1[(A - S)\delta]$$

over all such rules satisfying

$$\mathbb{P}_0(\delta = 1) \leq \alpha, \quad \mathbb{P}_1(\delta = 1) \geq 1 - \beta.$$

Below the feasibility threshold, we extend the capacity by zero as a reporting convention; the no-deployment rule does not satisfy the power constraint. The corresponding expected post-certification value, measured in economic units, is

$$\text{CAC}_{\alpha, \beta}(A) = \frac{\sigma^2}{2\gamma} c_{\alpha, \beta}(A).$$

3.1. Exact Gaussian certification threshold

Theorem 3.1 (Exact Gaussian certification threshold). *In the canonical Brownian likelihood experiment, the largest possible probability of deployment by information time A among all level- α rules is*

$$\pi_\alpha(A) = \Phi\left(\sqrt{A} - z_{1-\alpha}\right).$$

Consequently a false-deployment target α and power target $1 - \beta$ are statistically feasible by death if and only if

$$A \geq A_{\text{crit}}(\alpha, \beta) = (z_{1-\alpha} + z_{1-\beta})^2.$$

Equivalently, in KL units,

$$\mathcal{I}_{\text{life}} \geq \mathcal{I}_{\text{crit}}(\alpha, \beta) := \frac{1}{2}(z_{1-\alpha} + z_{1-\beta})^2.$$

Positive post-certification value requires $A > A_{\text{crit}}$: at the critical point the full lifetime is needed merely to attain the requested statistical discrimination.

Proof. If $A = 0$, the two observation laws coincide. No level- α decision has power exceeding α , and the decision $\mathbf{1}_{\{U_{\text{rnd}} \leq \alpha\}}$ attains it. Necessarily $S = 0$, so residual value is zero. Thus the displayed formula holds at $A = 0$. Assume $A > 0$ for the Gaussian threshold argument below. Every decision made by a rule with $S \leq A$ is $\mathcal{G}_A$ -measurable, so the Neyman–Pearson lemma applies to the full experiment observed up to A . The full path likelihood ratio at time A is e^{Z_A} , and Z_A is sufficient for testing the two simple path laws. Under $\mathbb{P}_0$ and $\mathbb{P}_1$,

$$Z_A \sim N(-A/2, A), \quad Z_A \sim N(A/2, A),$$

respectively. Neyman–Pearson gives the level- α threshold

$$c_A := -\frac{A}{2} + \sqrt{A} z_{1-\alpha}, \quad Z_A \geq c_A,$$

whose $\mathbb{P}_1$ -probability equals $\Phi(\sqrt{A} - z_{1-\alpha})$. Solving $\pi_\alpha(A) \geq 1 - \beta$ gives the displayed threshold. If $A > A_{\text{crit}}$, choose a deterministic test time $t \in (A_{\text{crit}}, A)$; the level- α Neyman–Pearson test at t has power greater than $1 - \beta$ and leaves positive time $A - t$ whenever it deploys. At the critical point with $A_{\text{crit}} > 0$, feasibility forces the rule to attain the extremal Neyman–Pearson receiver-operating-characteristic (ROC) point. Because the terminal likelihood ratio has a continuous distribution, the level- α most-powerful test is unique up to null sets; thus $\delta = \mathbf{1}_{E_A}$ almost surely, where $E_A := \{Z_A \geq c_A\}$. If $\mathbb{P}_1(S < A) > 0$, then on $\{S < A\}$ the conditional increment $Z_A - Z_S$ is Gaussian with strictly positive variance $A - S$, so $0 < \mathbb{P}_1(E_A \mid \mathcal{G}_S) < 1$. Since δ is $\mathcal{G}_S$ -measurable and $\delta = \mathbf{1}_{E_A}$, this would imply $\mathbb{P}_1(E_A \mid \mathcal{G}_S) = \delta \in \{0, 1\}$, a contradiction. Thus $S = A$ almost surely at the critical point and the post-certification value is zero. $\square$

For example, with $\alpha = 0.05$ and power 0.90,

$$A_{\text{crit}} = 8.5638473507\dots, \quad \mathcal{I}_{\text{crit}} = 4.2819236753\dots \text{ nats}.$$

If $\mathcal{I}$ denotes the KL information available to an arbitrary binary decision experiment, the generic data-processing bound is $\mathcal{I} \geq \text{kl}(0.9, 0.05) = 2.3762054\dots \text{ nats}$, which is below the canonical Gaussian requirement.

A constructive fixed-time lower bound for $A > A_{\text{crit}}$ is

$$c_{\alpha, \beta}(A) \geq \sup_{A_{\text{crit}} \leq t < A} (A - t) \Phi(\sqrt{t} - z_{1-\alpha}).$$

It tests once at time t . The optimal stopping rule can only improve on this benchmark. The zero residual value at the critical horizon is the endpoint implication of the most-powerful Gaussian test. In the discrete Gaussian experiment, Clerico et al. [9, Sections 6 and 8] likewise show that the hard-deadline power-optimal test cannot reject before its deadline. The next result quantifies the residual value obtainable when the information horizon exceeds that endpoint by a vanishing amount.

3.2. Near-critical capacity

Assume $0 < \alpha < 1 - \beta < 1$. For the remainder of the paper write $A_* := A_{\text{crit}}(\alpha, \beta)$. Let

$$\varphi_N(x) := (2\pi)^{-1/2} e^{-x^2/2}$$

denote the standard normal density. The fixed-time rule at A_* earns $(1 - \beta)(A - A_*)$ when it is embedded in a longer horizon. Sequential certification has a different local scale.

Theorem 3.2 (Sharp near-critical capacity). *For the canonical Brownian likelihood experiment,*

$$c_{\alpha,\beta}(A_*) = 0, \quad c_{\alpha,\beta}(A_* + \varepsilon) \sim \frac{\mathcal{K}_{\alpha,\beta}}{\log(1/\varepsilon)}, \quad \varepsilon \downarrow 0,$$

where

$$\mathcal{K}_{\alpha,\beta} := \frac{A_*}{2} \left([1 + z_{1-\beta}^2](1 - \beta) + z_{1-\beta} \varphi_N(z_{1-\beta}) \right). \quad (3.1)$$

The upper bound holds over randomized stopping rules. A binary conditional-error crossing rule attains the limit. Consequently

$$\lim_{\varepsilon \downarrow 0} \frac{c_{\alpha,\beta}(A_* + \varepsilon)}{\varepsilon} = +\infty.$$

The constant has a direct Gaussian interpretation. For a standard normal dummy variable N_G ,

$$\mathcal{K}_{\alpha,\beta} = \frac{A_*}{2} \mathbb{E}[(z_{1-\beta} - N_G)_+^2].$$

The positive part measures the squared terminal likelihood margin on paths that can support an earlier certificate; the expectation averages that margin across the alternative distribution. The constant measures information time in the canonical experiment; cash value follows from the payoff maps of Section 6.

For $\alpha = 0.05$ and power 0.90,

$$\mathcal{K}_{0.05,0.10} = 11.1460491583 \dots$$

Appendix E proves the universal upper bound and the attaining construction. The fixed-time benchmark is $(1 - \beta)\varepsilon$, so the adaptive-to-fixed-time value ratio diverges at rate

$$\frac{1}{\varepsilon \log(1/\varepsilon)}.$$

ε	Interim fraction t/A	Two-look value	Fixed-time value
10^{-1}	0.50	1.585621	9×10^{-2}
10^{-2}	0.80	1.001639	9×10^{-3}
10^{-3}	0.80	0.788752	9×10^{-4}
10^{-4}	0.80	0.610641	9×10^{-5}
10^{-5}	0.80	0.466293	9×10^{-6}
10^{-6}	0.90	0.386198	9×10^{-7}

Table 1: Residual information-time value at $\alpha = 0.05$, $\beta = 0.10$ and $A = A_* + \varepsilon$. The two-look column reports feasible policies selected from five interim fractions; it is a lower bound on unrestricted capacity. The fixed-time column tests at A_* .

The sharp law concerns a deterministic information horizon and allows the preterminal monitoring date to move with ε . Fixed monitoring grids and positive implementation leads define different local problems; Appendix E.5 records the corresponding boundary shift.

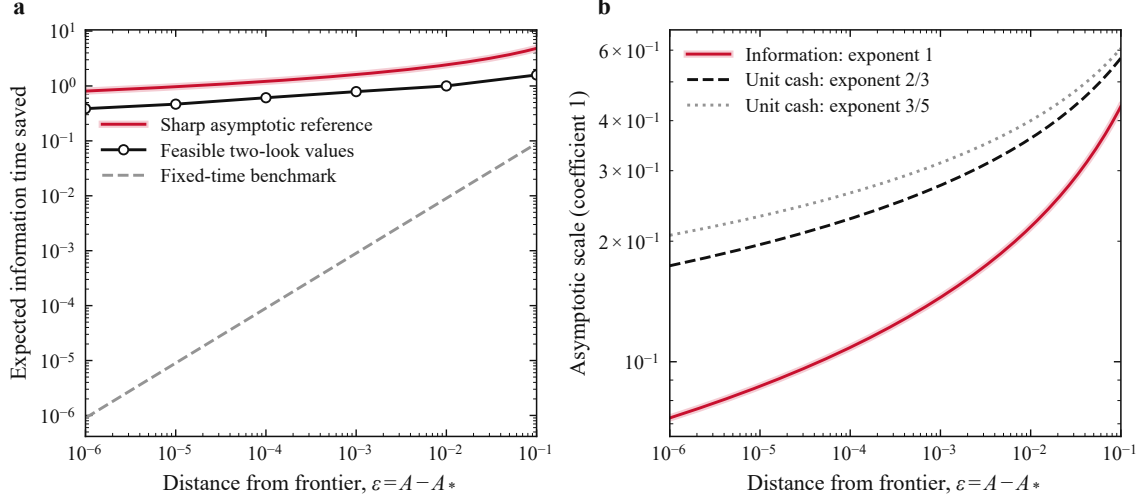


Figure 1: Near-frontier information and cash-value scales. Left: the sharp asymptotic reference $\mathcal{K}_{\alpha,\beta}/\log(1/\varepsilon)$, feasible two-look values from Table 1, and the fixed-time benchmark, at $\alpha = 0.05$ and $\beta = 0.10$. The reference is not a computed finite-horizon optimum. Right: logarithmic scales with coefficients normalized to one; the cash exponents require the uniform reward assumptions in Section 6. The exponent $(p+1)/(2p+1)$ is $2/3$ for a transverse crossing ($p = 1$) and $3/5$ for a quadratic tangency ($p = 2$).

3.3. A general information lower bound

Proposition 3.3 (Information lower bound for reliable certification). *Suppose $A > 0$ and $1 - \beta \geq \alpha$. Whenever the reliable constraint set is nonempty, its capacity satisfies*

$$\frac{A - c_{\alpha,\beta}(A)}{2} \geq \text{kl}(1 - \beta, \alpha).$$

Hence

$$\frac{\text{CAC}_{\alpha,\beta}(A)}{V^{PI}} \leq 1 - \frac{\text{kl}(1 - \beta, \alpha)}{\mathcal{I}_{\text{life}}}.$$

Feasibility requires the right-hand side to be nonnegative; otherwise no rule satisfies the declared power and type-I targets.

Proof. Take any feasible rule and write $R = \mathbb{E}_1[(A - S)\delta]$, $p = \mathbb{P}_1(\delta = 1)$, and $q = \mathbb{P}_0(\delta = 1)$. Setting $S = A$ on nondeployment gives

$$R = A - \mathbb{E}_1 S.$$

The stopped likelihood ratio is e^{Z_S} , so data processing from the stopped path to the binary decision δ yields

$$\frac{1}{2} \mathbb{E}_1 S = D_{\text{KL}}(\mathbb{P}_1^S \| \mathbb{P}_0^S) \geq \text{kl}(p, q) \geq \text{kl}(1 - \beta, \alpha),$$

where the last inequality uses $p \geq 1 - \beta \geq \alpha \geq q$ and the monotonicity of binary relative entropy on this region. Hence every feasible rule satisfies $R \leq A - 2 \text{kl}(1 - \beta, \alpha)$. Taking the supremum over feasible rules proves the result. $\square$

The exact Gaussian threshold gives the feasibility boundary in the canonical experiment. The KL inequality extends the information-budget argument beyond that model and also bounds the economic value consumed by certification.

4. Economic boundary and the value of information

Value-of-information methods evaluate evidence through the decision it changes [31–34]. Let $b_t < \theta_t$ be a predictable economic-boundary loading and impose the joint-likelihood and Girsanov assumptions of Section 2.1 for the gap $\theta - b$. The initial observation and predictor add no separate likelihood factor. For the corresponding laws $\mathbb{P}_\theta, \mathbb{P}_b$,

$$A_t^{\theta,b} = \sigma^{-2} \int_0^t (\theta_u - b_u)^2 X_u^2 du, \quad D_{\text{KL}}(\mathbb{P}_\theta^t \| \mathbb{P}_b^t) = \frac{1}{2} \mathbb{E}_\theta A_t^{\theta,b}.$$

Remaining flat during learning forgoes $\theta_t^2 X_t^2 dt / (2\gamma)$, so its price per KL nat is $\Pi_t^{\text{flat}} = (\sigma^2/\gamma)[\theta_t/(\theta_t - b_t)]^2$, which diverges near the boundary. The statistical experiment itself remains regular.

Theorem 4.1 (Value relative to the economic boundary). *At a fixed time, for $u(q; \theta) = q\theta X - \gamma q^2/2$ with $\gamma > 0$, the optima are $q_\theta = \theta X/\gamma$ and $q_b = bX/\gamma$. Their regret under the true loading is*

$$[u(q_\theta; \theta) - u(q_b; \theta)]dt = \frac{(\theta - b)^2 X^2}{2\gamma} dt = \frac{\sigma^2}{2\gamma} dA_t^{\theta,b}.$$

The boundary-relative price per KL nat is therefore $\Pi^{\text{relative}} = \sigma^2/\gamma$, independent of the gap.

Proof. Complete the square in u and divide the regret rate by the Gaussian drift-change KL rate $(\theta - b)^2 X^2 / (2\sigma^2)$. $\square$

The conversion depends on the admissible pre-certification policy: Π^{flat} applies to zero exposure, while the analysis below uses the boundary-relative benchmark.

4.1. Several assets: value per unit of information

For $dY_t = \mu dt + \Sigma^{1/2} dB_t$ and $u(q; \mu) = q^\top \mu - q^\top \Gamma q/2$, assume $\Sigma, \Gamma \in \mathbb{S}_{++}^d$. The true and boundary path laws share Σ ; put $\Delta = \mu - b$ and $\Psi = \Sigma^{1/2} \Gamma^{-1} \Sigma^{1/2}$.

Theorem 4.2 (Multivariate value–information bound). *For the boundary position $q_b = \Gamma^{-1}b$ relative to $q_\mu = \Gamma^{-1}\mu$, the regret and KL rates are $dR = \Delta^\top \Gamma^{-1} \Delta dt/2$ and $d\mathcal{I} = \Delta^\top \Sigma^{-1} \Delta dt/2$. For $\Delta \neq 0$,*

$$\frac{dR}{d\mathcal{I}} = \frac{\Delta^\top \Gamma^{-1} \Delta}{\Delta^\top \Sigma^{-1} \Delta} \in [\lambda_{\min}(\Psi), \lambda_{\max}(\Psi)].$$

The ratio is direction-independent exactly when $\Psi = cI_d$ for $c > 0$, equivalently $\Gamma = c^{-1}\Sigma$; then it equals c .

Proof. Complete the square for regret and use Gaussian Girsanov for KL. With $x = \Sigma^{-1/2}\Delta$, the ratio is the Rayleigh quotient $x^\top \Psi x / (x^\top x)$; it is constant in every direction exactly when $\Psi = cI_d$. $\square$

5. Alpha Survival Frontiers

Alpha decay has been studied both as a post-discovery empirical phenomenon and as an input to trading and strategy-durability models [35–38, 43]. The object here is the amount of statistical information that remains before the signal reaches its economic boundary. For the local terminal asymptotics in this section, take the drift and covariance paths to be deterministic; the same formulas hold pathwise on any realization for which the stated expansions are valid. Let the

observed d -dimensional return channel, under the true market model and an economic-boundary model, differ by drift gap

$$\Delta_t = \mu_t - b_t \in \mathbb{R}^d,$$

with instantaneous noise covariance $\Sigma_t \in \mathbb{S}_{++}^d$. Throughout this section, a superscript (b) on an information quantity means that KL divergence is measured relative to the economic-boundary model $b = (b_t)$. Let T be the end of the opportunity's economic lifetime. Assume that as $t \uparrow T$, for some $p > 0$ and nonzero $c \in \mathbb{R}^d$,

$$\Delta_t = c(T - t)^p + o((T - t)^p), \quad \Sigma_t \rightarrow \Sigma_T \succ 0.$$

Define remaining KL information from time t until death by

$$\mathcal{I}_{\text{rem}}^{(b)}(t) := \frac{1}{2} \int_t^T \Delta_s^\top \Sigma_s^{-1} \Delta_s ds.$$

Theorem 5.1 (Remaining information near the economic endpoint). *Under the assumptions above,*

$$\mathcal{I}_{\text{rem}}^{(b)}(t) = \frac{c^\top \Sigma_T^{-1} c}{2(2p + 1)} (T - t)^{2p+1} + o((T - t)^{2p+1}).$$

In particular, for a transverse crossing $p = 1$,

$$\mathcal{I}_{\text{rem}}^{(b)}(t) \sim \frac{c^\top \Sigma_T^{-1} c}{6} (T - t)^3.$$

Proof. Put $u = T - s$. By the assumed expansion and continuity of matrix inversion on the positive-definite cone,

$$\Delta_{T-u}^\top \Sigma_{T-u}^{-1} \Delta_{T-u} = u^{2p} c^\top \Sigma_T^{-1} c + o(u^{2p}).$$

Integrating from $u = 0$ to $u = T - t$ gives

$$\mathcal{I}_{\text{rem}}^{(b)}(t) = \frac{1}{2} c^\top \Sigma_T^{-1} c \frac{(T - t)^{2p+1}}{2p + 1} + o((T - t)^{2p+1}).$$

□

Economic value and the latest start. Put $V_{\text{rem}}^{\mu;b}(t) := \frac{1}{2} \int_t^T \Delta_s^\top \Gamma_s^{-1} \Delta_s ds$. For quadratic economic curvature $\Gamma_t \rightarrow \Gamma_T \succ 0$, completing the square and integrating as above gives $V_{\text{rem}}^{\mu;b}(t) \sim [c^\top \Gamma_T^{-1} c / (2(2p + 1))] (T - t)^{2p+1}$. Thus $V_{\text{rem}}^{\mu;b} / \mathcal{I}_{\text{rem}}^{(b)} \rightarrow (c^\top \Gamma_T^{-1} c) / (c^\top \Sigma_T^{-1} c)$. If $\Delta_{T-u} = cu^p$ and Σ is constant on the terminal interval, the information law is exact. With $\mathcal{I}_* = [z_{1-\alpha} + z_{1-\beta}]^2 / 2$, a newly started experiment needs at least

$$\ell_* = [2(2p + 1) \mathcal{I}_* / (c^\top \Sigma^{-1} c)]^{1/(2p+1)}$$

units of remaining time, provided this boundary lies in the stated interval. For a smooth crossing this gives the local asymptotic scale of the latest-start boundary; the global boundary depends on the full path geometry. An experiment already running also carries its likelihood state and remaining error budget.

Positive economic hurdle. For $\mu_t = \mu_0 e^{-\lambda_{\text{dec}} t}$, $0 < b < \mu_0$, $\lambda_{\text{dec}} > 0$, death occurs at $T = \lambda_{\text{dec}}^{-1} \log(\mu_0 / b)$. Writing $r_b = b / \mu_0$, direct integration against boundary drift b gives

$$\mathcal{I}_{\text{life}}^{(b)} = \frac{\mu_0^2}{2\sigma^2 \lambda_{\text{dec}}} \left[\frac{1}{2} - 2r_b + \frac{3}{2} r_b^2 + r_b^2 \log(1/r_b) \right] \sim \frac{\mu_0^2 (1 - r_b)^3}{6\sigma^2 \lambda_{\text{dec}}} \quad (r_b \uparrow 1).$$

The bracket expands as $(1 - r_b)^3 / 3 + O((1 - r_b)^4)$. Locally $\mu_{T-u} - b \sim b \lambda_{\text{dec}} u$, so Theorem 5.1 also gives $\mathcal{I}_{\text{rem}}^{(b)}(T - \ell) \sim b^2 \lambda_{\text{dec}}^2 \ell^3 / (6\sigma^2)$.

5.1. Zero-hurdle exponential model: the Alpha Survival Frontier

If the economic boundary is zero and $\mu_t = \mu_0 e^{-\lambda_{\text{dec}} t}$ for all $t \geq 0$, the calendar horizon is unbounded while total lifetime information remains finite:

$$\mathcal{I}_{\text{life}} = \frac{1}{2\sigma^2} \int_0^\infty \mu_0^2 e^{-2\lambda_{\text{dec}} t} dt = \frac{\mu_0^2}{4\sigma^2 \lambda_{\text{dec}}}.$$

Let

$$\mathbf{S} := \frac{\mu_0}{\sigma}, \quad h_{1/2} := \frac{\log 2}{\lambda_{\text{dec}}}$$

be the instantaneous Sharpe (with time units fixed consistently) and alpha half-life. Then

$$\mathcal{I}_{\text{life}} = \frac{\mathbf{S}^2 h_{1/2}}{4 \log 2}.$$

Reliable certification is feasible only if

$$\frac{\mathbf{S}^2 h_{1/2}}{4 \log 2} \geq \frac{1}{2} [z_{1-\alpha} + z_{1-\beta}]^2,$$

or equivalently

$$h_{1/2} \geq h_{\min}(\mathbf{S}; \alpha, \beta) := \frac{2 \log 2}{\mathbf{S}^2} [z_{1-\alpha} + z_{1-\beta}]^2.$$

We refer to this boundary as the canonical *Alpha Survival Frontier*.

For $\alpha = 0.05$, power $1 - \beta = 0.90$:

instantaneous Sharpe $\mathbf{S}$	minimum half-life
1.0	11.872
1.5	5.276
2.0	2.968
3.0	1.319

These thresholds use the calendar-time unit implicit in the scaling of $\mathbf{S}$ and are implied by the stated exponential-decay model.

5.2. Multiple testing and search breadth

Large-scale strategy search creates the familiar multiplicity and selection problem in empirical finance [12–14]. Suppose $M \in \mathbb{N}$ candidate signals are screened under familywise false-deployment budget $\alpha_F \in (0, 1)$. Under Bonferroni, assign

$$\alpha_M = \frac{\alpha_F}{M}.$$

The canonical information threshold for one selected signal is then

$$\mathcal{I}_{\text{crit}}^{(M)} = \frac{1}{2} [z_{1-\alpha_F/M} + z_{1-\beta}]^2.$$

Under a Bonferroni allocation, this is the exact per-test Gaussian threshold. Bonferroni remains valid under arbitrary dependence; additional dependence information can support less conservative allocations.

The zero-hurdle exponential survival frontier becomes

$$h_{\min}^{(M)}(\mathbf{S}) = \frac{2 \log 2}{\mathbf{S}^2} [z_{1-\alpha_F/M} + z_{1-\beta}]^2.$$

At $\alpha_F = 0.05$, power 0.90:

M	$h_{\min}, S = 1$	$S = 1.5$	$S = 2$	$S = 3$
1	11.872	5.276	2.968	1.319
10	20.627	9.168	5.157	2.292
100	28.979	12.880	7.245	3.220
1000	37.085	16.482	9.271	4.121

Search breadth and unequal budgets. Inverting the one-candidate power envelope gives the minimum level $\alpha_{\min}(\mathcal{I}, \beta) = \bar{\Phi}(\sqrt{2\mathcal{I}} - z_{1-\beta})$. Hence $M_{\max}^{\text{Bonf}} = \lfloor \alpha_F / \alpha_{\min} \rfloor$ (zero means no feasible candidate). Independent tests permit the Sidak inversion $M_{\max}^{\text{Sidak}} = \lfloor \log(1 - \alpha_F) / \log(1 - \alpha_{\min}) \rfloor$; Bonferroni needs no independence.

Remark 5.2 (Heterogeneous Bonferroni feasibility). Within the separable union-bound architecture, candidate j with information $\mathcal{I}_j \geq 0$ and power $1 - \beta_j$ needs $\alpha_j^* = \bar{\Phi}(\sqrt{2\mathcal{I}_j} - z_{1-\beta_j})$. All targets are feasible exactly when $\sum_j \alpha_j^* \leq \alpha_F$: necessity follows from the marginal envelopes and sufficiency from the union bound. With equal candidate values, sorting these costs and taking the longest feasible prefix maximizes cardinality; heterogeneous values give a 0–1 knapsack constraint.

For fixed α_F, β , Gaussian tail inversion gives $\mathcal{I}_{\text{crit}}^{(M)} \sim \log M$ as $M \rightarrow \infty$.

5.3. A dimensionless feasibility ratio

For a declared model, boundary and error allocation, define $\text{CR} = \mathcal{I}_{\text{life}} / \mathcal{I}_{\text{crit}}(\alpha, \beta)$, using the declared per-test level under multiplicity. In the canonical experiment, $\text{CR} < 1$ is infeasible, $\text{CR} = 1$ is terminal feasibility, and $\text{CR} > 1$ permits reliable certification. This ratio is a design-feasibility diagnostic; statistical evidence and deployment decisions depend on the observed likelihood and the selected stopping rule. An ongoing experiment must retain its likelihood state Z_s and original error allocation, as in Appendix B. The terminal envelope $\pi_\alpha(\text{CRA}_*)$ bounds the power of every rule satisfying the declared false-deployment bound; a stopping policy that preserves economic value can have lower power.

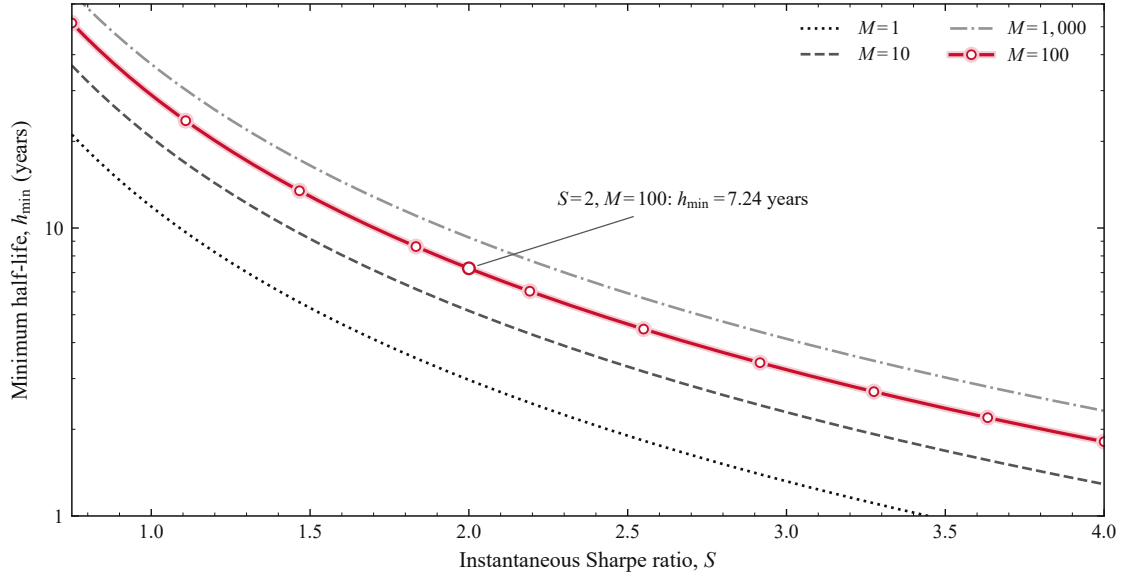


Figure 2: Canonical Alpha Survival Frontier for familywise false-deployment level 0.05 and power 0.90. Each curve gives the minimum half-life required at a given instantaneous Sharpe for a Bonferroni search over M candidates. The highlighted point is $S = 2$, $M = 100$, for which the minimum half-life is 7.245 years. The pale-red outline is a visual emphasis of the $M = 100$ curve.

6. Certification timing and post-certification cash value

The capacity $c_{\alpha,\beta}(A)$ measures information time left at certification. A trading payoff is generally a function of that residual clock. The sharp near-critical law from Section 3.2 transfers to payoff maps through their local dependence on residual information.

For $\rho_R > 0$ define the canonical residual-moment function

$$\mathcal{M}_{\alpha,\beta}(\rho_R) := A_*^{\rho_R} \mathbb{E}[(N_G + z_{1-\beta})_+^{2\rho_R}], \quad N_G \sim N(0, 1), \quad (6.1)$$

where $A_* = A_{\text{crit}}(\alpha, \beta)$ as in Section 3.2. The reward extension is taken along a family of canonical experiments with $A = A_* + \varepsilon \downarrow A_*$. Let $\mathcal{R}_A : [0, A] \rightarrow [0, \infty)$ be deterministic measurable residual rewards satisfying $\mathcal{R}_A(0) = 0$, uniformly bounded for A in a right neighborhood of A_* . Assume that, for some $a_0 > 0$ and $\rho_R > 0$,

$$\mathcal{R}_A(u)/u^{\rho_R} \longrightarrow a_0 \quad \text{uniformly as } A \downarrow A_* \text{ and } u \downarrow 0.$$

Appendix E.4 then gives

$$\sup_{(S,\delta)} \mathbb{E}_1[\mathcal{R}_A(A - S)\delta] \sim \frac{a_0 \mathcal{M}_{\alpha,\beta}(\rho_R)}{[2 \log(1/\varepsilon)]^{\rho_R}}, \quad \varepsilon \downarrow 0, \quad (6.2)$$

where the supremum is over the same reliable certification rules as in Section 3.

6.1. Bounded exposure near economic death

Use the scalar boundary-relative drift gap from Section 5. Suppose that, as $u \downarrow 0$,

$$\Delta_{T-u} = c u^p + o(u^p), \quad \sigma_{T-u} \rightarrow \sigma_T > 0, \quad c > 0,$$

with the same crossing order $p > 0$ as in Theorem 5.1. A unit position is opened only after certification and then held through the economic endpoint T . Its expected boundary-relative cash reward from time t is

$$V_{\text{unit}}(t) := \int_t^T \Delta_s ds.$$

Theorem 5.1 gives the residual information-time scale

$$2\mathcal{I}_{\text{rem}}^{(b)}(T - u) \sim \frac{c^2}{\sigma_T^2(2p + 1)} u^{2p+1},$$

while

$$V_{\text{unit}}(T - u) \sim \frac{c}{p + 1} u^{p+1}.$$

Hence

$$V_{\text{unit}} \sim K_p^{\text{unit}} u_{\text{info}}^{\rho_p}, \quad \rho_p := \frac{p + 1}{2p + 1}, \quad K_p^{\text{unit}} := \frac{c}{p + 1} \left(\frac{\sigma_T^2(2p + 1)}{c^2} \right)^{\rho_p}, \quad (6.3)$$

when $u_{\text{info}} := 2\mathcal{I}_{\text{rem}}^{(b)}$ denotes residual boundary-relative information time. For the near-critical statement, consider a family with total information $A \downarrow A_*$ whose terminal expansions are uniform and whose residual reward maps satisfy the hypotheses above. The constants c and σ_T denote positive limiting terminal coefficients. Combining (6.2) and (6.3) gives the sharp cash-value capacity

$$\text{CAC}_{\alpha,\beta}^{\text{cash}}(A_* + \varepsilon) \sim \frac{K_p^{\text{unit}} \mathcal{M}_{\alpha,\beta}(\rho_p)}{[2 \log(1/\varepsilon)]^{\rho_p}}. \quad (6.4)$$

For a transverse crossing, $p = 1$ and $\rho_p = 2/3$, so the attainable post-certification cash value is of order

$$[\log(1/\varepsilon)]^{-2/3}.$$

The exponent differs from the information-time exponent because a bounded position accumulates $\int \Delta_t dt$, whereas the likelihood clock accumulates $\int \Delta_t^2 / \sigma_t^2 dt$.

For the exponential gap

$$\Delta_t = \mu_0 e^{-\lambda_{\text{dec}} t} - b, \quad T = \lambda_{\text{dec}}^{-1} \log(\mu_0/b),$$

Section 5 has $p = 1$ and $c = b\lambda_{\text{dec}}$. Formula (6.4) applies directly.

6.2. A fixed deployment fee

A fixed round-trip fee shifts the last economically useful certification time away from the zero-edge endpoint. Consider the local linear gap

$$\Delta_t = c(T - t), \quad 0 \leq t \leq T,$$

and a fixed fee $F_{\text{fee}} \in (0, cT^2/2)$. The last entry time with nonnegative expected net cash reward is

$$T_{\text{fee}} = T - \ell_{\text{fee}}, \quad \ell_{\text{fee}} = \sqrt{\frac{2F_{\text{fee}}}{c}}.$$

Take constant volatility $\sigma > 0$ and use the information horizon ending at the last admissible entry date:

$$A_{\text{fee}} := \frac{1}{\sigma^2} \int_0^{T_{\text{fee}}} c^2(T - t)^2 dt = \frac{c^2}{3\sigma^2} (T^3 - \ell_{\text{fee}}^3).$$

The near-critical fee law below uses $A_{\text{fee}} = A_* + \varepsilon$. If u_{fee} is the information time remaining to T_{fee} , the exact net reward from certifying at that state is

$$\mathcal{R}_{\text{fee}}(u_{\text{fee}}) = \frac{c}{2} \left[\left(\ell_{\text{fee}}^3 + \frac{3\sigma^2 u_{\text{fee}}}{c^2} \right)^{2/3} - \ell_{\text{fee}}^2 \right]. \quad (6.5)$$

Thus

$$\mathcal{R}_{\text{fee}}(u_{\text{fee}}) \sim \frac{\sigma^2}{c\ell_{\text{fee}}} u_{\text{fee}}, \quad u_{\text{fee}} \downarrow 0,$$

and the sharp net-return capacity returns to the inverse-logarithmic scale:

$$\text{CAC}_{\alpha,\beta}^{\text{fee}}(A_* + \varepsilon) \sim \frac{\sigma^2 \mathcal{K}_{\alpha,\beta}}{c\ell_{\text{fee}} \log(1/\varepsilon)}. \quad (6.6)$$

The fee changes the local asymptotic law because the relevant deployment boundary is reached while the underlying drift gap is still strictly positive.

6.3. Cash value beyond lifetime information

The ratio CR identifies statistical feasibility in the canonical simple-hypothesis experiment. Cash value also depends on how the reward is distributed along the information clock. In particular, two opportunities can share the same current edge, volatility, calendar horizon, total future cash value, lifetime information and the same CR , while having different local crossing orders at economic death. Equation (6.4) then gives different certification-constrained cash values. Appendix E.6 gives an explicit pair with identical CR and different asymptotic exponents.

7. Unknown decay and robust certification

The preceding sections condition on a specified boundary-relative drift profile. We now require a single observable policy to satisfy the same false-deployment and power targets over a declared family of possible decay paths.

7.1. Ordered families

Fix a deterministic horizon T and let $\{\Delta_{\vartheta} : \vartheta \in \Theta\}$ be continuous scalar drift gaps satisfying

$$\Delta_{\vartheta}(t) \geq \underline{\Delta}(t) > 0, \quad 0 \leq t < T, \quad \underline{\Delta}(T) \geq 0,$$

where the least profile $\underline{\Delta}$ belongs to the family. Under alternative ϑ ,

$$dY_t = \Delta_{\vartheta}(t) dt + \sigma dB_t,$$

while the boundary law has zero drift. Define the least-profile information clock and matched log-likelihood by

$$\begin{aligned} \underline{A}_t &:= \frac{1}{\sigma^2} \int_0^t \underline{\Delta}(u)^2 du, & \underline{A} &:= \underline{A}_T, \\ \underline{Z}_t &:= \frac{1}{\sigma^2} \int_0^t \underline{\Delta}(u) dY_u - \frac{1}{2} \underline{A}_t. \end{aligned}$$

For $0 < \alpha < 1 - \beta < 1$, define the robust residual-information capacity by

$$c_{\alpha,\beta}^{\text{rob}}(\underline{A}) := \sup_{(S,\delta)} \inf_{\vartheta \in \Theta} \mathbb{E}_{\vartheta}[(\underline{A} - S)\delta],$$

where $S \leq \underline{A}$ is a stopping time on the deterministic least-profile information clock, $0 \leq \delta \leq 1$ is measurable at S , and the common policy satisfies $\mathbb{E}_0 \delta \leq \alpha$ and $\inf_{\vartheta} \mathbb{E}_{\vartheta} \delta \geq 1 - \beta$. The declared family is suppressed in the capacity notation.

Theorem 7.1 (Ordered-family robust frontier). *Among all terminal decisions of size at most α ,*

$$\sup_{\delta: \mathbb{E}_0 \delta \leq \alpha} \inf_{\vartheta \in \Theta} \mathbb{E}_{\vartheta} \delta = \Phi\left(\sqrt{\underline{A}} - z_{1-\alpha}\right).$$

Consequently a common certificate with power at least $1 - \beta$ for every $\vartheta \in \Theta$ exists if and only if

$$\underline{A} \geq A_*.$$

If $\underline{A} = A_ + \varepsilon$, the robust residual-information capacity satisfies*

$$c_{\alpha, \beta}^{\text{rob}}(\underline{A}) \sim \frac{\mathcal{K}_{\alpha, \beta}}{\log(1/\varepsilon)}.$$

For a common bounded, nonnegative, nondecreasing reward $\mathcal{R}(u)$ with $\mathcal{R}(0) = 0$, assume $\mathcal{R}(u) \sim a_0 u^{\rho_{\text{R}}}$ with the uniformity required by the reward-transfer result. Then $\sup_{(S, \delta)} \inf_{\vartheta} \mathbb{E}_{\vartheta}[\mathcal{R}(\underline{A} - S)\delta]$ obeys (6.2) with the least-profile constant. The same conclusion holds for member-specific payoff maps when every member's reward at each calendar time dominates this least-profile reward.

The matched statistic is ordered pathwise under a common Brownian coupling, and the least profile belongs to the declared family. It determines terminal worst-case power. The conditional-error crossing rule stops no later for larger profiles; under the stated reward-ordering assumption, this also establishes the sharp robust-reward lower bound. The least member supplies the matching upper bound.

For the exponential family

$$\Delta_{\theta, \lambda_{\text{dec}}}(t) = \theta e^{-\lambda_{\text{dec}} t} - b,$$

with $\theta \in [\theta_-, \theta_+]$, $\lambda_{\text{dec}} \in [\lambda_{\text{dec}, -}, \lambda_{\text{dec}, +}]$ and $\theta_- > b > 0$, the common least profile up to the earliest economic death is

$$\underline{\Delta}(t) = \theta_- e^{-\lambda_{\text{dec}, +} t} - b, \quad T = \lambda_{\text{dec}, +}^{-1} \log(\theta_- / b).$$

The policy is constructed from $\underline{\Delta}$ and the observed path, uniformly over the declared parameter rectangle.

7.2. Nonordered families and least-favorable mixtures

For nonordered profiles, robust feasibility depends on the geometry of the alternative family beyond memberwise lifetime information. Consider a finite Gaussian experiment with whitened observation $\mathbf{Y} \in \mathbb{R}^n$:

$$\mathbf{Y} \sim N(0, I_n) \quad (\mathbb{P}_0), \quad \mathbf{Y} \sim N(m_j, I_n) \quad (\mathbb{P}_j), \quad j = 1, \dots, J.$$

For a terminal decision $0 \leq \delta \leq 1$, define

$$\pi_{\text{rob}} := \sup_{\mathbb{E}_0 \delta \leq \alpha} \min_{1 \leq j \leq J} \mathbb{E}_j \delta.$$

Let

$$L_j(y) = \exp\left(m_j^\top y - \frac{1}{2} \|m_j\|^2\right),$$

and let

$$\mathcal{W}_J := \left\{ w \in [0, 1]^J : \sum_{j=1}^J w_j = 1 \right\}$$

be the simplex of mixture weights. For $w = (w_1, \dots, w_J) \in \mathcal{W}_J$, set $L_w = \sum_j w_j L_j$.

Proposition 7.2 (Least-favorable mixture). *The exact robust terminal power satisfies*

$$\pi_{\text{rob}} = \min_{w \in \mathcal{W}_J} \sup_{0 \leq \delta \leq 1, \mathbb{E}_0 \delta \leq \alpha} \mathbb{E}_0[L_w \delta].$$

For fixed w , the inner problem is the Neyman–Pearson test of P_0 against the mixture likelihood L_w . A saddle pair exists.

This formulation exposes a strict uncertainty penalty at the individual frontier. If two distinct nonzero alternatives satisfy

$$\|m_1\|^2 = \|m_2\|^2 = A_*,$$

then each member separately attains power $1 - \beta$ at size α , but no common level- α decision attains that power under both members.

For the two exponential profiles used in the numerical benchmark, calibrated so that each profile individually lies at $CR = 1$ for $\alpha = 0.05$ and power 0.90, the optimal common test has worst-profile power

$$0.8655488924 \dots$$

Scaling both profile norms by the same factor, uniform power 0.90 is recovered at

$$CR = 1.1268349125 \dots$$

Thus the geometry of the alternative family enters robust certification even when every profile has the same oracle lifetime information.

7.3. Transfer from an estimated model

Let Π be a common class of admissible policies. For $\pi \in \Pi$, let $\delta_\pi \in \{0, 1\}$ denote its deployment decision and let $R_j(\pi)$ denote its integrable reward under alternative j . Write

$$\text{size}(\pi) = P_0(\delta_\pi = 1), \quad \text{pow}_j(\pi) = P_j(\delta_\pi = 1), \quad \mathcal{V}_j(\pi) = \mathbb{E}_j[R_j(\pi)].$$

Approximate-model quantities carry hats. Suppose uniformly over Π and the declared alternatives that

$$\begin{aligned} |\text{size}(\pi) - \widehat{\text{size}}(\pi)| &\leq \eta_P, \\ |\text{pow}_j(\pi) - \widehat{\text{pow}}_j(\pi)| &\leq \eta_P, \quad |\mathcal{V}_j(\pi) - \widehat{\mathcal{V}}_j(\pi)| \leq \eta_V. \end{aligned}$$

Fix the scalar objective before optimization. Here we use

$$\mathcal{U}(\pi) := \inf_j \mathcal{V}_j(\pi), \quad \widehat{\mathcal{U}}(\pi) := \inf_j \widehat{\mathcal{V}}_j(\pi).$$

A policy optimized for $\widehat{\mathcal{U}}$ under the tightened approximate constraints

$$\widehat{\text{size}} \leq \alpha - \eta_P, \quad \widehat{\text{pow}}_j \geq 1 - \beta + \eta_P \quad \text{for all } j,$$

satisfies the original reliability contract under the true model. Its worst-case value gap is bounded by the optimization error, $2\eta_V$, and the value of the statistical safety margin, provided the twice-tightened true feasible set is nonempty. This is a bound for the declared scalar objective, not simultaneous optimality for every memberwise reward. Appendix F gives the precise statement. For an estimated model, the stated uniform approximation bounds must hold conditionally on the information used to freeze that model, or on an explicitly specified event; they do not follow merely from fitting a parametric family.

8. Market equilibrium with certification costs

Information acquisition and arbitrage are endogenous in several market-equilibrium models [21, 23–25]. More recent work studies competition among factor investors and the empirical role of post-publication attenuation, arbitrage capital, and implementation frictions [26–30]. The preceding sections take opportunity lifetime as given. We now allow arbitrage activity to shorten it. An explicit crowding map links aggregate research intensity to the remaining information budget, so the CAC threshold becomes an equilibrium constraint.

Retain the reliable-information threshold

$$\mathcal{I}_* := \mathcal{I}_{\text{crit}}(\alpha, \beta) = \frac{1}{2} [z_{1-\alpha} + z_{1-\beta}]^2.$$

Let $v_{\text{KL}} > 0$ denote economic value per KL nat. In the canonical scalar model, Theorem 4.1 gives $v_{\text{KL}} = \sigma^2/\gamma$. Define the optimal reliable CAC economic surplus at lifetime information $\mathcal{I} \geq 0$ by

$$C^*(\mathcal{I}) := \frac{v_{\text{KL}}}{2} c_{\alpha, \beta}(2\mathcal{I}).$$

The equilibrium arguments below use only the resulting value function and its stated regularity; Appendix B gives the corresponding randomized constrained-stopping representation. For the equilibrium result we require the natural regularity that C^* is continuous, with

$$C^*(\mathcal{I}) = 0 \quad (\mathcal{I} \leq \mathcal{I}_*), \quad C^*(\mathcal{I}) > 0 \quad (\mathcal{I} > \mathcal{I}_*).$$

In the canonical model, the CAC value is strictly increasing above the frontier. Indeed, if one KL nat of perfect-information value is worth v_{KL} , then embedding any ε -optimal rule at $\mathcal{I}_1$ in the longer horizon $\mathcal{I}_2 > \mathcal{I}_1 \geq \mathcal{I}_*$ adds $v_{\text{KL}}(\mathcal{I}_2 - \mathcal{I}_1)$ on every deployment, whose probability under the alternative law is at least $1 - \beta$. Letting $\varepsilon \downarrow 0$ gives

$$C^*(\mathcal{I}_2) - C^*(\mathcal{I}_1) \geq v_{\text{KL}}(1 - \beta)(\mathcal{I}_2 - \mathcal{I}_1).$$

Thus the constructive linear surplus

$$C_0(\mathcal{I}) = v_{\text{KL}}(1 - \beta)(\mathcal{I} - \mathcal{I}_*)_+$$

is a lower envelope of the optimal CAC value.

Let $\mathcal{I}_0 > 0$ be the no-crowding lifetime-information endowment and let $x \geq 0$ denote aggregate arbitrage/research intensity. The remaining information budget is

$$\mathcal{I} = G(x),$$

where $G : [0, \infty) \rightarrow [0, \mathcal{I}_0]$ is continuous and strictly decreasing, with $G(0) = \mathcal{I}_0$. The canonical exponential-crowding benchmark uses a crowding sensitivity $\chi > 0$ and is

$$G(x) = \frac{\mathcal{I}_0}{1 + \chi x},$$

which arises when $\lambda_{\text{dec},0} > 0$ denotes the baseline exponential decay rate and crowding accelerates it from $\lambda_{\text{dec},0}$ to $\lambda_{\text{dec},0}(1 + \chi x)$.

Suppose each unit of active arbitrage capital pays marginal research or implementation cost $\psi > 0$. In a competitive market with many individually small arbitrageurs, aggregate post-certification value $C^*(G(x))$ is allocated proportionally across active capacity. Positive-entry equilibrium satisfies

$$\frac{C^*(G(x))}{x} = \psi,$$

or equivalently

$$C^*(G(x)) = \psi x.$$

Theorem 8.1 (Competitive equilibrium under certification costs). *Assume C^* and G have the properties above. If $\mathcal{I}_0 > \mathcal{I}_*$ and there is a finite x_* with $G(x_*) = \mathcal{I}_*$, then for every $\psi > 0$ there is a unique positive equilibrium $x_\psi \in (0, x_*)$ satisfying*

$$C^*(G(x_\psi)) = \psi x_\psi.$$

Under the entry dynamic

$$\dot{x} = \eta\{C^*(G(x)) - \psi x\}, \quad \eta > 0,$$

this equilibrium is globally asymptotically stable. Its residual information $\mathcal{I}_\psi = G(x_\psi)$ satisfies

$$\mathcal{I}_* < \mathcal{I}_\psi < \mathcal{I}_0, \quad \mathcal{I}_\psi \downarrow \mathcal{I}_* \quad \text{as } \psi \downarrow 0.$$

If $\mathcal{I}_0 \leq \mathcal{I}_$, the no-entry equilibrium $x = 0$ is globally stable.*

Proof. Put $F_\psi(x) = C^*(G(x)) - \psi x$. It is continuous and strictly decreasing, with $F_\psi(0) > 0$ and $F_\psi(x_*) = -\psi x_* < 0$, so it has one root in $(0, x_*)$. Its sign points toward that root on $[0, \infty)$; monotonicity implies the one-sided Lipschitz inequality $(x - y)(F_\psi(x) - F_\psi(y)) \leq 0$, hence uniqueness of the scalar flow and global asymptotic stability. For $0 < \psi_1 < \psi_2$, evaluating F_{ψ_1} at x_{ψ_2} shows $x_{\psi_1} > x_{\psi_2}$. Its limit as $\psi \downarrow 0$ cannot be below x_* , since then $C^*(G(x_\psi))$ would have a positive limit while $\psi x_\psi \rightarrow 0$. Thus $x_\psi \uparrow x_*$ and $\mathcal{I}_\psi \downarrow \mathcal{I}_*$. If $\mathcal{I}_0 \leq \mathcal{I}_*$, the flow is $\dot{x} = -\eta\psi x$. $\square$

The equilibrium divides opportunities at the certification frontier. Below the frontier, reliable post-certification value is zero. Above the frontier, rents finance entry and crowding reduces the remaining information budget. As research cost falls, the equilibrium residual information approaches the reliable-information threshold.

The lower envelope C_0 also yields a useful nonasymptotic bound. Since G is strictly decreasing, write G^{-1} for its inverse on its range. The optimal equilibrium satisfies

$$\mathcal{I}_\psi - \mathcal{I}_* \leq \frac{\psi G^{-1}(\mathcal{I}_*)}{v_{\text{KL}}(1 - \beta)}.$$

For $G(x) = \mathcal{I}_0/(1 + \chi x)$,

$$\mathcal{I}_\psi - \mathcal{I}_* \leq \frac{\psi}{\chi v_{\text{KL}}(1 - \beta)} \left(\frac{\mathcal{I}_0}{\mathcal{I}_*} - 1 \right).$$

Hence the residual excess information above the CAC frontier vanishes at least linearly with research cost, without requiring differentiability of the stopping value at the threshold.

Corollary 8.2 (Sharp concentration of the equilibrium gap). *Under the hypotheses of Theorem 8.1, assume $0 < \alpha < 1 - \beta < 1$ and retain*

$$C^*(\mathcal{I}) = \frac{v_{\text{KL}}}{2} c_{\alpha, \beta}(2\mathcal{I}).$$

Let $x_ = G^{-1}(\mathcal{I}_*)$. As $\psi \downarrow 0$,*

$$\log \frac{1}{\mathcal{I}_\psi - \mathcal{I}_*} \sim \frac{v_{\text{KL}} \mathcal{K}_{\alpha, \beta}}{2x_*} \frac{1}{\psi}. \quad (8.1)$$

Equivalently,

$$\mathcal{I}_\psi - \mathcal{I}_* = \exp \left[-\frac{v_{\text{KL}} \mathcal{K}_{\alpha, \beta}}{2x_* \psi} + o(1/\psi) \right].$$

Proof. Theorem 8.1 gives $x_\psi \rightarrow x_* > 0$ and $\mathcal{I}_\psi \downarrow \mathcal{I}_*$. Since $A_* = 2\mathcal{I}_*$, Theorem 3.2 gives

$$c_{\alpha,\beta}(2\mathcal{I}_\psi) \sim \frac{\mathcal{K}_{\alpha,\beta}}{\log(1/[2(\mathcal{I}_\psi - \mathcal{I}_*)])}.$$

The equilibrium identity $C^*(\mathcal{I}_\psi) = \psi x_\psi$ implies

$$\log\left(\frac{1}{2(\mathcal{I}_\psi - \mathcal{I}_*)}\right) \sim \frac{v_{\text{KL}}\mathcal{K}_{\alpha,\beta}}{2x_*\psi}.$$

The additive constant $\log 2$ is negligible on the $1/\psi$ scale, which proves (8.1). $\square$

8.1. A finite number of competing arbitrageurs

The proportional allocation in the large-number competitive model can itself be obtained as the limit of a strategic game with finitely many arbitrageurs. Let $N \geq 2$ arbitrage desks choose efforts $e_i \geq 0$, with aggregate $E := \sum_{i=1}^N e_i$; write e_{-i} for the vector of efforts of desks other than i . Let

$$R(E) := C^*(G(E)).$$

For $E > 0$, desk i receives proportional share e_i/E of the endogenous post-certification value pool and pays ψe_i :

$$u_i(e_i, e_{-i}) = \frac{e_i}{E} R(E) - \psi e_i, \quad u_i(0, \mathbf{0}) := 0.$$

At a symmetric interior equilibrium, write $E_N > 0$ for aggregate equilibrium effort. Differentiability of R at E_N gives

$$\psi = \left(1 - \frac{1}{N}\right) \frac{R(E_N)}{E_N} + \frac{1}{N} R'(E_N).$$

The second term is a strategic prize-destruction effect. If $R'(E_N) < 0$, a finite arbitrageur internalizes the reduction in the total opportunity caused by additional effort. For the following comparison, assume symmetric interior equilibria exist for every $N \geq 2$, that R is continuously differentiable and strictly decreasing in its positive-rent region $(0, E_*)$, with $R(0) > 0$ and $R(E_*) = 0$, and that

$$F_N(E) := \left(1 - \frac{1}{N}\right) \frac{R(E)}{E} + \frac{1}{N} R'(E)$$

is strictly decreasing there. Then the corresponding roots E_N increase with N , since $F_{N+1}(E) > F_N(E)$, and converge to the unique competitive root $R(E)/E = \psi$ for $\psi > 0$. These are conditional comparative statics; the first-order equation alone does not establish existence of a Nash equilibrium.

A closed-form benchmark is obtained by using the globally defined clipped linear information impact

$$G(E) = (\mathcal{I}_0 - \kappa_{\text{crowd}} E)_+, \quad \kappa_{\text{crowd}} > 0,$$

with the constructive post-certification value

$$C_0(\mathcal{I}) = a(\mathcal{I} - \mathcal{I}_*)_+, \quad a = v_{\text{KL}}(1 - \beta).$$

Proposition 8.3 (Exact finite- N linear benchmark). *Let $N \geq 2$, $\mathcal{I}_0 > \mathcal{I}_*$, $a, \kappa_{\text{crowd}} > 0$, and $\psi \geq 0$. Writing $\Delta\mathcal{I} = \mathcal{I}_0 - \mathcal{I}_*$, define the rent pool for all $E \geq 0$ by $R(E) = a(\Delta\mathcal{I} - \kappa_{\text{crowd}} E)_+$. The proportional-share game has a unique symmetric equilibrium with positive gross rent. If*

$\psi > 0$, it is also the unique symmetric equilibrium with positive aggregate effort. Its aggregate effort is

$$E_N = \frac{a\Delta\mathcal{I}(N-1)}{N(a\kappa_{\text{crowd}} + \psi)}.$$

Define the equilibrium residual information by $\mathcal{I}_N^{\text{eq}} := \mathcal{I}_0 - \kappa_{\text{crowd}}E_N$. The equilibrium remains in the positive-rent region, and its residual information satisfies

$$\mathcal{I}_N^{\text{eq}} - \mathcal{I}_* = \Delta\mathcal{I} \frac{a\kappa_{\text{crowd}} + N\psi}{N(a\kappa_{\text{crowd}} + \psi)}.$$

Equivalently,

$$\frac{\mathcal{I}_N^{\text{eq}} - \mathcal{I}_*}{\mathcal{I}_0 - \mathcal{I}_*} = \frac{\psi}{a\kappa_{\text{crowd}} + \psi} + \frac{a\kappa_{\text{crowd}}}{N(a\kappa_{\text{crowd}} + \psi)}.$$

In particular, if $\psi = 0$, then

$$\mathcal{I}_N^{\text{eq}} = \mathcal{I}_* + \frac{\mathcal{I}_0 - \mathcal{I}_*}{N}.$$

Proof. See Appendix B.1. □

The first term in the residual-information fraction comes from research cost; the second comes from having only finitely many competitors. Along the symmetric positive-rent equilibrium branch, a finite set of arbitrageurs leaves positive excess information even when research itself is costless; only the large-number competitive limit of this branch reaches the CAC frontier. In the symmetric linear benchmark, $1/N$ is also the Herfindahl index of effort shares, so the zero-cost residual-information fraction equals market concentration. With nonlinear crowding or asymmetric payoffs the relationship changes with the market technology.

In the zero-hurdle exponential Gaussian model, $\mathcal{I} = S^2 h_{1/2} / (4 \log 2)$. Consequently, for opportunities that are actively compressed by many low-cost arbitrageurs,

$$S^2 h_{1/2} \longrightarrow 4 \log 2 \mathcal{I}_*.$$

This limit describes opportunities whose lifetime is actively compressed by many low-cost arbitrageurs.

Remark 8.4. The equilibrium comparisons condition on implementation technology. Heterogeneity in research cost, crowding and competitive breadth can change their pooled sign; the opportunity's information budget alone is not a cross-sectional prediction of its rate of decay [30].

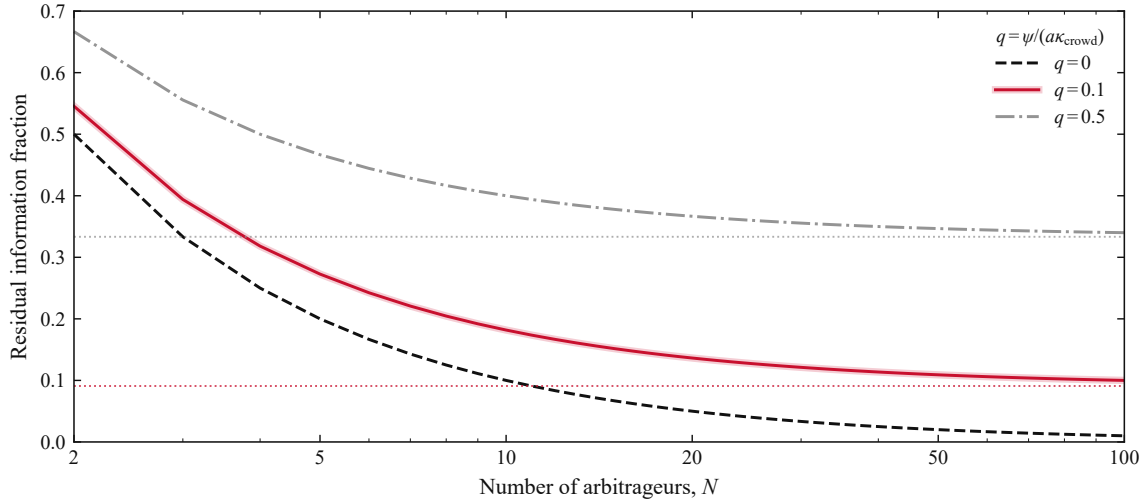


Figure 3: Residual excess lifetime information along the symmetric positive-rent equilibrium branch of the exact linear-information strategic benchmark. With zero research cost the residual fraction is exactly $1/N$; positive research cost creates an additional limiting gap even as the number of competitors becomes large. Dotted lines mark $q/(1+q)$, where $q = \psi/(a\kappa_{\text{crowd}})$; the pale-red outline highlights $q = 0.1$.

9. Funding persistence as a lifetime signal

Funding persistence provides a descriptive illustration [39–42]. The benchmark uses BTC-USDT and ETH-USDT on Binance, Bybit, Gate, HTX, KuCoin and MEXC at eight-hour settlements. Events require 180 pre-event days, absolute funding above its trailing 95th percentile, a seven-day cooldown and complete seven-day follow-up; 543 of 545 events have finite scores. The $W = 540$ pre-event settlements fit the no-intercept AR(1) $f_{j+1} = \phi f_j + \varepsilon_{j+1}$ by least squares, with residual sample standard deviation $\hat{\sigma}_\varepsilon$. For $0 < \hat{\phi} < 1$, set $\hat{\lambda}_{\text{dec}} = -\log \hat{\phi}$ and $L_{\text{score}} = f_t^2 / [4\hat{\sigma}_\varepsilon^2 \hat{\lambda}_{\text{dec}} \mathcal{I}_{\text{crit}}(0.05, 0.10)]$. The threshold is 4.2819236753... nats; normalization leaves ranks unchanged. Comparators are $z_{\text{edge},t} = |f_t|/s_{f,t}$, where $s_{f,t}$ is the pre-event funding standard deviation, and the historical t statistic of $\text{sgn}(f_t)f_j$. The outcome is $Y_{7d,t} = \sum_{k=1}^{21} \text{sgn}(f_t)f_{t+k}$. Figure 4 displays the estimates computed from the supplied event table: lifetime, edge and historical- t associations are 0.4741, 0.3833 and 0.3606. Residualizing lifetime midranks on edge midranks with an intercept leaves association 0.2777 (95% interval [0.1228, 0.4061]). Intervals use 5,000 calendar-month cluster bootstrap resamples across 38 months, with seed 1729. On the same 543-event panel, the rank correlation between L_{score} and the sign-oriented AR(1) cumulative forecast $\hat{m}_t = |f_t| \sum_{k=1}^{21} \hat{\phi}^k$ is 0.9222, rising to 0.9952 when this forecast is divided by its conditional AR(1) standard deviation. The reproduction guide supplies the forecast-variance formula, event table and source identities. The score thus summarizes the AR(1) persistence channel. These retrospective quantities concern signed funding; net trading returns additionally require execution, hedge, financing and collateral costs.

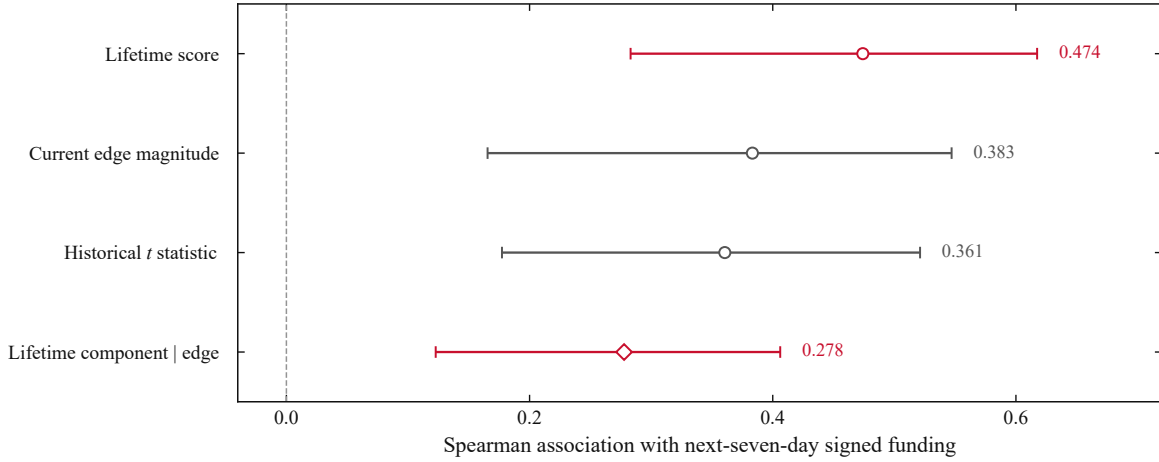


Figure 4: Descriptive funding benchmark, 543 events. Points report Spearman associations with next-seven-day sign-oriented funding; horizontal whiskers show calendar-month cluster bootstrap 95% intervals from 5,000 resamples. The last row residualizes lifetime-score rank on current-edge rank.

10. Discussion

CAC distinguishes evidence already accumulated, evidence that can still arrive before economic death, and value remaining after certification. A research team can use the frontier to screen a proposed validation budget: a high current edge does not compensate for an information lifetime too short to satisfy the declared reliability contract. The screen concerns a new experiment; an experiment already in progress also requires its current likelihood state.

For a portfolio manager, residual information and implementable cash value are separate objects. Near the frontier, their logarithmic rates can differ, and equal certifiability ratios need not imply equal economic rankings. Trading limits, fees, implementation delays and the shape of the remaining drift must enter the reward map. The theorems concern hypotheses within a specified drift experiment; realized portfolio profits lie outside their scope.

The theoretical conclusions also have explicit model limits. The canonical likelihood must describe the joint observed experiment; predictability of a signal alone does not make its likelihood contribution disappear. Reward transfer requires uniform local behavior and control away from the boundary. Ordered uncertainty preserves a least-profile frontier only with a compatible reward order; nonordered alternatives require a common minimax test. Estimated-model guarantees require uniform approximation bounds that include the null.

The funding exercise connects the lifetime perspective to an observable market quantity. Its retrospective associations concern signed funding and same-sign persistence. They do not identify the incremental return of a deployable trading strategy: that estimand would require a common execution contract, financing and portfolio constraints, and a separate prospective comparison. The article's empirical claim is confined to the persistence channel; it does not claim confirmatory validation of trading profitability.

11. Conclusion

This perspective suggests treating validation speed as part of strategy design. A useful research rule must account for both the evidence needed to act and the opportunity remaining when that evidence arrives. Extending the analysis to richer observation models, implementable portfolio

constraints and endogenous information acquisition would connect the certification frontier more closely to the allocation of arbitrage capital.

12. Code and data availability

The AlphaValue research package contains the numerical routines, synthetic examples, tests, editable figures and reproduction code for the theoretical and descriptive funding results reported here. The software archive is versioned at Zenodo under the concept DOI [10.5281/zenodo.23070742](https://doi.org/10.5281/zenodo.23070742). Third-party funding histories remain subject to their source licenses; the reproduction guide identifies the required source revision and input paths. The derived event table, source-file identities and numerical references are included. The reproduction command recomputes the funding estimates and intervals offline; reconstructing the event table from raw histories requires the pinned inputs. The public bundle is scoped to this article. A clean editable installation uses `python -m pip install -e '[test,plot]'` followed by `python reproduce/run.py`.

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A. Capacity with a type-I error constraint

With only a size constraint, let $c_\alpha(A) = \sup \mathbb{E}_1[(A - S)\delta]$ over $S \in \mathcal{T}_A$ and binary $\mathcal{G}_S$ -measurable δ with $\mathbb{P}_0(\delta = 1) \leq \alpha$. Independent randomization is available at time zero, so $c_\alpha(A) \geq \alpha A$. The economic value is $\text{CAC}_\alpha(A) = \sigma^2 c_\alpha(A)/(2\gamma)$.

Theorem A.1 (Nonasymptotic converse without a fixed power target). *For $A > 0$, $r_\alpha = c_\alpha(A)/A$ obeys $A(1 - r_\alpha)/2 \geq \text{kl}(r_\alpha, \alpha)$, and is bounded above by the unique root $\bar{r} \in [\alpha, 1)$ of $A(1 - \bar{r}) = 2 \text{kl}(\bar{r}, \alpha)$.*

Proof. For a feasible rule of reward R , set $r = R/A$, $p = \mathbb{P}_1(\delta = 1)$, $q = \mathbb{P}_0(\delta = 1)$ and $S = A$ on nondeployment. Then $p \geq r$, $q \leq \alpha$, and $(A - R)/2 = D_{\text{KL}}(\mathbb{P}_1^S \parallel \mathbb{P}_0^S) \geq \text{kl}(p, q) \geq \text{kl}(r, \alpha)$ when $r \geq \alpha$. Apply this to a maximizing sequence and use continuity; the case $r_\alpha = \alpha$ is immediate. The root is unique because $A(1 - r) - 2 \text{kl}(r, \alpha)$ strictly decreases from positive to negative on $[\alpha, 1)$. $\square$

As $A \downarrow 0$, $r_\alpha \rightarrow \alpha$: the limiting value comes from the permitted lottery alone. An evidence-added report can subtract αA .

Constructive threshold. For $h > 0$, put $\tau_h = \inf\{s : Z_s \geq h\}$. Brownian drift μ has first-passage cdf

$$F_\mu(t; h) = \Phi((\mu t - h)/\sqrt{t}) + e^{2\mu h} \Phi((- \mu t - h)/\sqrt{t}).$$

The unique h_A with $F_{-1/2}(A; h_A) = \alpha$ gives the exact-level value $L_\alpha(A) = \int_0^A F_{1/2}(t; h_A) dt \leq c_\alpha(A)$. The anytime boundary $h_\infty = \log(1/\alpha)$ has $\mathbb{P}_0(\tau_{h_\infty} < \infty) = \alpha$.

Theorem A.2 (Asymptotic information cost of certification). *For fixed $\alpha \in (0, 1)$, $A - c_\alpha(A) \rightarrow 2 \log(1/\alpha)$ as $A \rightarrow \infty$. Hence*

$$\frac{\text{CAC}_\alpha(A)}{V^{PI}} = 1 - \frac{\log(1/\alpha)}{\mathcal{I}_{\text{life}}} + o(\mathcal{I}_{\text{life}}^{-1}), \quad V^{PI} - \text{CAC}_\alpha(A) \rightarrow \frac{\sigma^2}{\gamma} \log(1/\alpha).$$

Proof. Under $\mathbb{P}_1$, the hitting time of $h = \log(1/\alpha)$ has mean $2h$, so $g_A := A - c_\alpha(A) \leq \mathbb{E}_1(\tau_h \wedge A) \rightarrow 2h$. The converse gives $g_A/2 \geq \text{kl}(1 - g_A/A, \alpha)$; boundedness of g_A makes its right side converge to h , proving the matching lower bound. $\square$

B. Constrained stopping representation

Setting $S = A$ on nondeployment makes the objective $A - \mathbb{E}_1 S$, linking it to optimal group-sequential design [5, 6, 8]. In the convexified class, a rule uses an $\mathcal{G}_S$ -measurable deployment probability $\rho_{\text{dep}} \in [0, 1]$. Write $\mathcal{T}_{s,A}$ for stopping times in $[s, A]$ and $\mathbb{E}_{s,z}^{(1)}$ for the likelihood diffusion started at $Z_s = z$.

Theorem B.1 (Lagrange representation for the type-I constrained problem). *In the randomized stopping class,*

$$c_\alpha(A) = \inf_{\lambda_{\text{Lag}} \geq 0} \{\lambda_{\text{Lag}} \alpha + V_{\lambda_{\text{Lag}}}(0, 0)\}, \quad V_{\lambda_{\text{Lag}}}(s, z) = \sup_{S \in \mathcal{T}_{s,A}} \mathbb{E}_{s,z}^{(1)}(A - S - \lambda_{\text{Lag}} e^{-Z_S})_+.$$

Proof. The stopped likelihood identity gives $\mathbb{E}_0 \rho_{\text{dep}} = \mathbb{E}_1[e^{-Z_S} \rho_{\text{dep}}]$. Maximizing the Lagrangian pointwise over $\rho_{\text{dep}} \in [0, 1]$ gives the positive-part payoff and weak duality. Randomized stopping times on a bounded horizon are compact in the Baxter–Chacon topology [44]; bounded rewards and uniformly integrable stopped likelihood ratios give continuity of reward and size in this

Brownian model. The attainable set is therefore compact and convex. The no-deploy rule strictly satisfies size when $\alpha > 0$, so separation gives a nonnegative multiplier and no duality gap. $\square$

This construction uses the classical constrained-stopping framework [15, 16], with related deadline formulations in [17–20]. With generator $\mathcal{L} = \frac{1}{2}\partial_{zz} + \frac{1}{2}\partial_z$, the continuation equation is $\partial_s V + \mathcal{L}V = 0$, with obstacle $(A - s - \lambda_{\text{Lag}} e^{-z})_+$. In the positive-obstacle region, $(\partial_s + \mathcal{L})(A - s - \lambda_{\text{Lag}} e^{-z}) = -1$, and the stopping boundary is an upper boundary $b_{\lambda_{\text{Lag}}}(s)$. For $\lambda_{\text{Lag}} > 0$, $V_{\lambda_{\text{Lag}}}(s, z) = V_1(s, z - \log \lambda_{\text{Lag}})$ and $b_{\lambda_{\text{Lag}}} = \log \lambda_{\text{Lag}} + b_1$. Under smooth fit and the standard boundary regularity, the early-exercise representation and boundary equation are

$$V_{\lambda_{\text{Lag}}}(s, z) = \int_s^A \mathbb{P}_{s,z}^{(1)}\{Z_u \geq b_{\lambda_{\text{Lag}}}(u)\} du,$$

$$A - s - \lambda_{\text{Lag}} e^{-b_{\lambda_{\text{Lag}}}(s)} = \int_s^A \Phi\left(\frac{b_{\lambda_{\text{Lag}}}(s) + (u - s)/2 - b_{\lambda_{\text{Lag}}}(u)}{\sqrt{u - s}}\right) du.$$

For reliable CAC, add $\omega \geq 0$: the stopping payoff becomes $(A - S + \omega - \lambda_{\text{Lag}} e^{-Z_S})_+$, defining $V_{\lambda_{\text{Lag}}, \omega}$, and the dual objective is $\lambda_{\text{Lag}} \alpha - \omega(1 - \beta) + V_{\lambda_{\text{Lag}}, \omega}(0, 0)$ under strong-duality hypotheses. Feasibility must first be checked by Theorem 3.1. For finite multipliers and $s < A$, waiting until any deterministic $u \in (s, A)$ has positive expected payoff by Gaussian full support. Immediate nondeployment cannot be optimal in this cost-free dual; an abandonment region requires observation cost or another binding constraint. The continuation state (s, z) includes both time and log likelihood.

B.1. Finite-player equilibrium proof

Proof of Proposition 8.3.

Proof. In the positive-rent region the payoff is

$$u_i = \frac{e_i}{E} a \Delta \mathcal{I} - (a \kappa_{\text{crowd}} + \psi) e_i.$$

Set $\mathcal{V}_0 := a \Delta \mathcal{I}$ and $k := a \kappa_{\text{crowd}} + \psi$. Define the aggregate rival effort by $E_{-i} := \sum_{j \neq i} e_j$. Against $E_{-i} > 0$, the first and second derivatives are

$$\frac{\partial u_i}{\partial e_i} = \frac{\mathcal{V}_0 E_{-i}}{(e_i + E_{-i})^2} - k, \quad \frac{\partial^2 u_i}{\partial e_i^2} = -\frac{2 \mathcal{V}_0 E_{-i}}{(e_i + E_{-i})^3} < 0.$$

At a symmetric equilibrium $e_i = e$ and $E = Ne$, the first-order condition gives $e = \mathcal{V}_0(N - 1)/(N^2 k)$ and hence the stated E_N . Moreover,

$$\frac{\kappa_{\text{crowd}} E_N}{\Delta \mathcal{I}} = \frac{a \kappa_{\text{crowd}} (N - 1)}{N(a \kappa_{\text{crowd}} + \psi)} < 1,$$

so the solution lies strictly above the CAC frontier. Substituting $\mathcal{I}_N^{\text{eq}} = \mathcal{I}_0 - \kappa_{\text{crowd}} E_N$ yields the residual-information identities. A unilateral deviation that pushes aggregate effort beyond the positive-rent region obtains zero gross rent and nonpositive net payoff, whereas the stated interior equilibrium gives a strictly positive payoff. Strict concavity in each player's own effort within the positive-rent region gives uniqueness among symmetric equilibria with positive gross rent. When $\psi > 0$, any zero-rent profile with positive symmetric effort yields negative payoff and is improved by choosing zero. When $\psi = 0$, additional zero-rent equilibria can exist, and no global uniqueness claim is made. $\square$

C. Unknown scale and predictable regressors

C.1. A pairwise KL necessary condition

Let $\mathfrak{P}_0, \mathfrak{P}_1$ be declared null and alternative laws on the observed sample space, and require $\sup_{Q_0 \in \mathfrak{P}_0} Q_0(\delta = 1) \leq \alpha$ and $\inf_{Q_1 \in \mathfrak{P}_1} Q_1(\delta = 1) \geq 1 - \beta > \alpha$.

Theorem C.1 (Pairwise KL necessary condition). *Every compatible pair $(Q_1, Q_0) \in \mathfrak{P}_1 \times \mathfrak{P}_0$ with $Q_1 \ll Q_0$ satisfies $D_{\text{KL}}(Q_1 \| Q_0) \geq \text{kl}(1 - \beta, \alpha)$.*

Proof. Data processing to δ gives the binary KL; on $p \geq 1 - \beta > \alpha \geq q$, it is minimized at $(1 - \beta, \alpha)$. $\square$

AR(1) benchmark. For $X_{t+1} = \phi X_t + \xi_{t+1}$, $Y_{t+1} = \theta X_t + \varepsilon_{t+1}$, assume independent centered Gaussian innovation sequences of variances $v_X, v_R > 0$, $|\phi| < 1$, and stationary X . The boundary companion has loading b and the same nuisance parameters and initial law. The expected per-observation KL is

$$\kappa_{\text{KL}}(\theta, \phi, v_R, v_X; b) = \frac{(\theta - b)^2 v_X}{2v_R(1 - \phi^2)}.$$

For a declared compact rectangle, let $\underline{\kappa}_{\text{KL}}$ be its infimum. The minimum uses the smallest $|\theta - b|$ and $|\phi|$, smallest v_X and largest v_R ; if the persistence interval contains zero, its minimum is at zero. Every minimizing boundary companion must belong to the null family. If $\underline{\kappa}_{\text{KL}} > 0$, then $L_{\text{hor}} \geq \lceil \text{kl}(1 - \beta, \alpha) / \underline{\kappa}_{\text{KL}} \rceil$ is necessary. Marginal envelopes without a compatible joint pair do not prove impossibility.

C.2. Sequential certification with predictable regressors

For $\mathcal{F}_t$ -measurable X_t before observing Y_{t+1} , assume $Y_{t+1} = \theta X_t + \varepsilon_{t+1}$, $\varepsilon_{t+1} \mid \mathcal{F}_t \sim N(0, v)$ with $v \leq \bar{v}$. Test $\theta \leq b$ against $\theta \geq b + d$, $d > 0$. For $q > 0$, define

$$\ell_\alpha = \log(1/\alpha), \quad \lambda_q = \sqrt{2\ell_\alpha / (\bar{v}q)}, \quad Q_t = \sum_{s < t} X_s^2, \quad U_t = \sum_{s < t} X_s(Y_{s+1} - bX_s),$$

and $E_t = \exp(\lambda_q U_t - \lambda_q^2 \bar{v} Q_t / 2)$, $\tau_q = \inf\{t : Q_t \geq q\}$, $\tau_E = \inf\{t : E_t \geq 1/\alpha\}$.

Theorem C.2 (Predictable-design e-process certificate). *Under every null, E_t is a nonnegative supermartingale and $\mathbb{P}_{\theta, v}(\tau_E < \infty) \leq \alpha$. Under every alternative, if $\mathcal{I}_q = d^2 q / (2\bar{v}) > \ell_\alpha$,*

$$\mathbb{P}_{\theta, v}(\tau_E > \tau_q, \tau_q < \infty) \leq e^{-(\sqrt{\mathcal{I}_q} - \sqrt{\ell_\alpha})^2}.$$

Consequently $\mathcal{I}_q \geq [\sqrt{\log(1/\alpha)} + \sqrt{\log(1/\beta)}]^2$ gives power $1 - \beta$ by τ_q if $\tau_q < \infty$ almost surely throughout the alternative class. Without reachability, only the displayed joint-event bound is asserted.

Proof. The conditional Gaussian mgf is at most one under the null, proving validity. Under an alternative, put $N_t = \sum_{s < t} X_s \varepsilon_{s+1}$; then $\log E_t \geq (\lambda_q d - \lambda_q^2 \bar{v} / 2) Q_t + \lambda_q N_t$. Failure at finite τ_q implies $N_{\tau_q} \leq -c_q Q_{\tau_q}$, where $c_q = d - \lambda_q \bar{v} / 2 - \ell_\alpha / (\lambda_q q) = d - \sqrt{2\ell_\alpha \bar{v} / q} > 0$. Optional sampling and Markov's inequality for the nonnegative supermartingale $\exp(-\zeta N_t - \zeta^2 \bar{v} Q_t / 2)$ bound this event by $\exp[-q(\zeta c_q - \zeta^2 \bar{v} / 2)]$. Optimizing at $\zeta = c_q / \bar{v}$ gives the result. $\square$

Estimated d or $\bar{v}$ require adding their confidence-set failure to the error budget. Unknown-variance inference is discussed in [10, 11]; the software includes fixed-design Studentization and Normal-Inverse-Gamma monitoring benchmarks, whose joint-model and clock assumptions must be checked separately.

C.3. Discrete information–value identity

Use the stationary AR model above over L_{hor} periods. A position q_t with $q_{-1} = 0$ earns $\theta X_t q_t - \gamma q_t^2/2 - \lambda_{\text{tr}}(q_t - q_{t-1})^2/2$, with $\gamma > 0$, $\lambda_{\text{tr}} \geq 0$, and terminal penalty $\eta_T q_{L_{\text{hor}}-1}^2/2$, $\eta_T \geq 0$. Define backward from $K_{L_{\text{hor}}} = \eta_T$, $\Lambda_{L_{\text{hor}}}^q = 0$,

$$\begin{aligned} D_t &= \gamma + \lambda_{\text{tr}} + K_{t+1}, & a_t &= \lambda_{\text{tr}}/D_t, & g_t(\phi) &= (1 + \phi \Lambda_{t+1}^q)/D_t, \\ K_t &= \lambda_{\text{tr}}(\gamma + K_{t+1})/D_t, & \Lambda_t^q &= \lambda_{\text{tr}} g_t(\phi). \end{aligned}$$

The full-information and boundary feedbacks are $q_t^\theta = a_t q_{t-1}^\theta + \theta g_t X_t$ and $q_t^b = a_t q_{t-1}^b + b g_t X_t$.

Theorem C.3 (Discrete information–value identity). *The expected regret of the boundary policy and return-channel information satisfy*

$$R_{L_{\text{hor}}}^{\theta:b} = \frac{v_X(\theta - b)^2}{2(1 - \phi^2)} \sum_{t=0}^{L_{\text{hor}}-1} D_t g_t^2 = \Pi_{L_{\text{hor}}} \mathcal{I}_{L_{\text{hor}}, \text{return}}^{\theta:b},$$

where $\mathcal{I}_{L_{\text{hor}}, \text{return}}^{\theta:b} = L_{\text{hor}} v_X(\theta - b)^2/[2v_R(1 - \phi^2)]$ and $\Pi_{L_{\text{hor}}} = (v_R/L_{\text{hor}}) \sum_t D_t g_t^2$. With $\lambda_{\text{tr}} = \eta_T = 0$, $\Pi_{L_{\text{hor}}} = v_R/\gamma$.

Proof. Backward completion of the Bellman square gives the feedbacks. Evaluated at the boundary policy’s own inventory, its one-period Bellman loss is $D_t[(\theta - b)g_t X_t]^2/2$. Continuation values telescope, including terminal inventory cost. Take stationary expectations $\mathbb{E}X_t^2 = v_X/(1 - \phi^2)$ and divide by KL. When costs vanish, $D_t g_t^2 = 1/\gamma$. $\square$

For contemporaneous innovation correlation $|\rho_{\text{corr}}| < 1$ with the same variances, joint Gaussian KL is $\mathcal{I}_{L_{\text{hor}}, \text{full}}^{\theta:b} = \mathcal{I}_{L_{\text{hor}}, \text{return}}^{\theta:b}/(1 - \rho_{\text{corr}}^2)$; the full-information price per nat is $(1 - \rho_{\text{corr}}^2)\Pi_{L_{\text{hor}}}$. Zero correlation supplies the least-information member only when it belongs to the declared family.

D. Random economic lifetime

Let $H = A_T$, and consider a $\mathbb{G}$ -stopping time $S \leq H$ and a binary $\mathcal{G}_S$ -measurable deployment decision δ , with no deployment at death. Assume $\mathbb{E}_1 H < \infty$, the stopped likelihood identity, and that observing the deadline adds no likelihood factor omitted from Z . Localization and L^2 optional sampling give $D_{\text{KL}}(\mathbb{P}_1^S \parallel \mathbb{P}_0^S) = \mathbb{E}_1 S/2$. Data processing therefore implies the necessary obstruction

$$\frac{1}{2} \mathbb{E}_1 H \geq \text{kl}(1 - \beta, \alpha)$$

for size α and power $1 - \beta > \alpha$. The mean is insufficient: lower tails control how often certification can finish. Let τ_{cert} be the deployment time, with $\tau_{\text{cert}} = +\infty$ on nondeployment. If $F(h)$ and $S_H(h)$ lower-bound $\mathbb{P}_1(\tau_{\text{cert}} \leq h)$ and $\mathbb{P}_1(H > h)$ for a fixed rule, the union bound gives $\mathbb{P}_1(\tau_{\text{cert}} < H) \geq \sup_h [F(h) + S_H(h) - 1]_+$ without independence. The canonical terminal power envelope gives the complementary upper bound $\mathbb{P}_1(\delta = 1, S < H) \leq \inf_h \{\pi_\alpha(h) + \mathbb{P}_1(H > h)\} \wedge 1$ whenever the stopped experiment embeds in that canonical experiment.

For a common deadline law independent of the likelihood process under both hypotheses, let $H \sim \text{Exp}(\varrho)$, $\varrho > 0$, and deploy at $\tau_h = \inf\{s : Z_s \geq h\}$ only if $\tau_h < H$. Put $s_\varrho = \sqrt{1/4 + 2\varrho}$, $a_0 = s_\varrho + 1/2$ and $a_1 = s_\varrho - 1/2$. The Brownian first-passage Laplace transform gives

$$\mathbb{P}_0(\tau_h < H) = e^{-a_0 h}, \quad \mathbb{P}_1(\tau_h < H) = e^{-a_1 h}.$$

Thus $h = -\log \alpha/a_0$ gives exact size α and power $\pi_{\alpha,\varrho}^{\text{exp}} = \alpha^{a_1/a_0}$. This constant-threshold rule meets the power target when that quantity is at least $1 - \beta$. Setting $r = \log(1 - \beta)/\log \alpha \in (0, 1)$ and solving equality yields

$$\varrho_{\text{bench}} = \frac{r}{2(1-r)^2}, \quad \frac{\mathbb{E}_1 H}{2} = \frac{(1-r)^2}{r}.$$

At $\alpha = 0.05$, power 0.90, the benchmark mean is about 26.47 KL nats versus 4.28 for a deterministic deadline. This constant-threshold rule gives a constructive benchmark; the optimal random-deadline frontier remains open.

E. Sharp near-critical capacity

This appendix proves Theorem 3.2 and the reward transfer used in Section 6. Throughout,

$$A_* := A_{\text{crit}}(\alpha, \beta) = (z_{1-\alpha} + z_{1-\beta})^2,$$

and

$$c_A := -\frac{A}{2} + \sqrt{A} z_{1-\alpha}, \quad \delta_A^{\text{NP}} := \mathbf{1}_{\{Z_A \geq c_A\}}, \quad \pi_\alpha(A) := \Phi(\sqrt{A} - z_{1-\alpha}).$$

A randomized rule consists of a stopping time $S \leq A$ and a deployment probability $0 \leq \delta \leq 1$ measurable with respect to $\mathcal{G}_S$, satisfying

$$\mathbb{E}_0 \delta \leq \alpha, \quad \mathbb{E}_1 \delta \geq 1 - \beta.$$

Write

$$u := A - S, \quad x := Z_S - c_A.$$

For $\rho_{\text{R}} > 0$, define the residual-moment capacity

$$c_{\alpha,\beta}^{[\rho_{\text{R}}]}(A) := \sup \mathbb{E}_1[u^{\rho_{\text{R}}} \delta]$$

over the same reliable rule class, and set

$$\mathcal{M}_{\alpha,\beta}(\rho_{\text{R}}) := A_*^{\rho_{\text{R}}} \mathbb{E} \left[(N_{\text{G}} + z_{1-\beta})_+^{2\rho_{\text{R}}} \right], \quad N_{\text{G}} \sim N(0, 1).$$

Theorem E.1 (Sharp residual-moment law). *For every $\rho_{\text{R}} > 0$,*

$$c_{\alpha,\beta}^{[\rho_{\text{R}}]}(A_*) = 0,$$

and

$$\lim_{\varepsilon \downarrow 0} [\log(1/\varepsilon)]^{\rho_{\text{R}}} c_{\alpha,\beta}^{[\rho_{\text{R}}]}(A_* + \varepsilon) = 2^{-\rho_{\text{R}}} \mathcal{M}_{\alpha,\beta}(\rho_{\text{R}}).$$

The upper bound holds over randomized rules, and a binary conditional-error crossing rule attains the limit.

Theorem 3.2 is the case $\rho_{\text{R}} = 1$.

E.1. Conditional Gaussian penalty

For $N_G \sim N(0, 1)$ define

$$\mathfrak{q}(u, x) := \mathbb{E} \left[(e^{-x-u/2-\sqrt{u}N_G} - 1)_+ \right].$$

Fix $A_{\max} > A_*$ and $\eta \in (0, 1)$. There exists $c_\eta > 0$ such that, for $0 < u \leq A_{\max}$,

$$\mathfrak{q}(u, x) \geq c_\eta u \exp \left[-\frac{(1+\eta)x_+^2}{2u} \right]. \quad (\text{E.1})$$

To obtain the bound, restrict N_G to

$$[-x_+/\sqrt{u} - \sqrt{u}/2 - 2, -x_+/\sqrt{u} - \sqrt{u}/2 - 1].$$

On this interval the payoff is at least $e^{\sqrt{u}} - 1 \geq \sqrt{u}$. A lower bound on the Gaussian density, together with $(a+b)^2 \leq (1+\eta)a^2 + (1+\eta^{-1})b^2$, gives (E.1).

For a feasible rule at $A = A_* + \varepsilon$, define the Neyman–Pearson deficit

$$\begin{aligned} \mathfrak{d}_A &:= \mathbb{E}_0[(e^{Z_A} - e^{c_A})(\delta_A^{\text{NP}} - \delta)] \\ &= \pi_\alpha(A) - \mathbb{E}_1\delta - e^{c_A}(\alpha - \mathbb{E}_0\delta). \end{aligned} \quad (\text{E.2})$$

The integrand is nonnegative pointwise, and hence

$$0 \leq \mathfrak{d}_A \leq \pi_\alpha(A) - (1 - \beta) = O(\varepsilon).$$

Conditioning at S and retaining the paths that deploy below the terminal Neyman–Pearson cutoff yields

$$\mathfrak{d}_A \geq \mathbb{E}_1[\delta \mathbf{1}_{\{u>0\}} \mathfrak{q}(u, x)]. \quad (\text{E.3})$$

Under $\mathbb{P}_1$, conditional on $\mathcal{G}_S$, the increment $Z_A - Z_S$ is $N(u/2, u)$.

E.2. Universal upper bound

Let

$$\Lambda_\varepsilon := \log(1/\varepsilon)$$

and define

$$\mathcal{B}_\eta := \left\{ u > 0 : u\Lambda_\varepsilon > \frac{1+\eta}{2(1-\eta)} x_+^2 \right\}.$$

On $\mathcal{B}_\eta$, (E.1) and (E.3) imply

$$\mathbb{E}_1[\delta u \mathbf{1}_{\mathcal{B}_\eta}] = O(\varepsilon^\eta).$$

If $0 < \rho_R \leq 1$, concavity on the measure with density $\delta \mathbf{1}_{\mathcal{B}_\eta}$ gives

$$\mathbb{E}_1[\delta u^{\rho_R} \mathbf{1}_{\mathcal{B}_\eta}] \leq \{\mathbb{E}_1[\delta u \mathbf{1}_{\mathcal{B}_\eta}]\}^{\rho_R}.$$

If $\rho_R \geq 1$, use $u^{\rho_R} \leq A_{\max}^{\rho_R-1} u$. Hence the contribution of $\mathcal{B}_\eta$, multiplied by $\Lambda_\varepsilon^{\rho_R}$, vanishes uniformly over feasible rules.

On $\mathcal{B}_\eta^c$,

$$\Lambda_\varepsilon^{\rho_R} u^{\rho_R} \leq \left(\frac{1+\eta}{2(1-\eta)} \right)^{\rho_R} x_+^{2\rho_R}.$$

Taking $\rho_R = 1$ first gives $\mathbb{E}_1[\delta u] = O(\Lambda_\varepsilon^{-1})$. Since δ is $\mathcal{G}_S$ -measurable,

$$\mathbb{E}_1[\delta(Z_A - Z_S)^2] = \mathbb{E}_1[\delta(u + u^2/4)] = O(\Lambda_\varepsilon^{-1}).$$

Conditional Gaussian moment bounds then yield, for every fixed $k > 0$,

$$\mathbb{E}_1[\delta|Z_A - Z_S|^k] \longrightarrow 0.$$

Consequently,

$$\mathbb{E}_1[\delta(Z_S - c_A)_+^{2\rho_R}] \leq \mathbb{E}_1[(Z_A - c_A)_+^{2\rho_R}] + o(1).$$

Letting $\eta \downarrow 0$ gives

$$\limsup_{\varepsilon \downarrow 0} \Lambda_\varepsilon^{\rho_R} c_{\alpha, \beta}^{[\rho_R]}(A_* + \varepsilon) \leq 2^{-\rho_R} \mathcal{M}_{\alpha, \beta}(\rho_R).$$

At $A = A_*$, (E.2) gives $\mathfrak{d}_A = 0$. Since $\mathfrak{q}(u, x) > 0$ for every $u > 0$ and finite x , (E.3) forces $\delta u = 0$ almost surely, so $c_{\alpha, \beta}^{[\rho_R]}(A_*) = 0$.

E.3. Attaining conditional-error crossing rule

For $A > A_*$ define the terminal threshold calibrated to power $1 - \beta$,

$$\tilde{c}_A := A/2 - \sqrt{A} z_{1-\beta},$$

and its null rejection probability

$$\alpha_A^{\text{term}} := \mathbb{P}_0(Z_A \geq \tilde{c}_A) = 1 - \Phi(\sqrt{A} - z_{1-\beta}) < \alpha, \quad \bar{\alpha}_A := \frac{\alpha_A^{\text{term}}}{\alpha}.$$

Under $\mathbb{P}_0$ the conditional terminal-rejection probability

$$C_t^{\text{ce}} := \mathbb{P}_0(Z_A \geq \tilde{c}_A \mid \mathcal{G}_t) = \Phi\left(\frac{Z_t - \tilde{c}_A - (A - t)/2}{\sqrt{A - t}}\right), \quad t < A,$$

is a bounded continuous martingale. Let S_A be its first crossing of $\bar{\alpha}_A$ before A , with $S_A = A$ if no crossing occurs, and set

$$\delta_A^{\text{ce}} := \mathbf{1}_{\{S_A < A\}}.$$

Every terminal-positive path crosses the boundary. Optional sampling gives

$$\mathbb{P}_0(\delta_A^{\text{ce}} = 1) = \alpha, \quad \mathbb{P}_1(\delta_A^{\text{ce}} = 1) \geq \mathbb{P}_1(Z_A \geq \tilde{c}_A) = 1 - \beta.$$

Thus the rule is feasible.

As $\varepsilon \downarrow 0$,

$$1 - \bar{\alpha}_A \sim \frac{\varphi_N(z_{1-\alpha})}{2\alpha\sqrt{A_*}} \varepsilon, \quad [\Phi^{-1}(\bar{\alpha}_A)]^2 \sim 2\Lambda_\varepsilon.$$

On a crossing, let

$$u_A^{\text{rem}} := A - S_A.$$

Then

$$\Phi^{-1}(\bar{\alpha}_A) \sqrt{u_A^{\text{rem}}} = Z_{S_A} - \tilde{c}_A - u_A^{\text{rem}}/2.$$

Couple the experiments under $\mathbb{P}_1$ using $Z_t = B_t + t/2$ on $[0, A_{\max}]$, and put $u_A^{\text{rem}} = 0$ when no crossing occurs. For all A sufficiently close to A_* , write $k_A = \Phi^{-1}(\bar{\alpha}_A) > 0$. There is a random variable M_{dom} with every finite moment, independent of A , such that on a crossing

$$0 \leq k_A \sqrt{u_A^{\text{rem}}} = Z_{S_A} - \tilde{c}_A - u_A^{\text{rem}}/2 \leq M_{\text{dom}}.$$

Thus $k_A^2 u_A^{\text{rem}} \leq M_{\text{dom}}^2$, and the residual time tends to zero on crossing paths. If $Z_{A_*} > c_{A_*}$, terminal rejection, and hence a crossing, occurs for all sufficiently small $A - A_*$. Brownian

continuity then gives convergence to $(Z_{A_*} - c_{A_*})^2$. If $Z_{A_*} < c_{A_*}$, an infinite sequence of crossings would have $S_A \rightarrow A_*$ and a negative limiting right-hand side in the crossing identity, which is impossible. The equality event has probability zero. Consequently,

$$k_A^2 u_A^{\text{rem}} \rightarrow (Z_{A_*} - c_{A_*})_+^2$$

almost surely and, by domination, in every finite L^q . Since $[\Phi^{-1}(\bar{\alpha}_A)]^2/(2\Lambda_\varepsilon) \rightarrow 1$,

$$\Lambda_\varepsilon^{\rho_R} \mathbb{E}_1[(u_A^{\text{rem}})^{\rho_R} \delta_A^{\text{ce}}] \rightarrow 2^{-\rho_R} \mathcal{M}_{\alpha, \beta}(\rho_R).$$

This matches the upper bound and proves Theorem E.1. At $\rho_R = 1$,

$$\mathcal{K}_{\alpha, \beta} = \frac{1}{2} \mathcal{M}_{\alpha, \beta}(1) = \frac{A_*}{2} \left([1 + z_{1-\beta}^2](1 - \beta) + z_{1-\beta} \varphi_N(z_{1-\beta}) \right).$$

E.4. General residual rewards

Let $\mathcal{R}_A : [0, A] \rightarrow [0, \infty)$ be measurable and uniformly bounded for A near A_* , with $\mathcal{R}_A(0) = 0$. Suppose $\mathcal{R}_A(u)/u^{\rho_R} \rightarrow a_0 > 0$ uniformly as $A \downarrow A_*$ and $u \downarrow 0$. Fix $h > 0$ and choose $m > \rho_R$. The residual-moment theorem gives, uniformly over feasible rules,

$$\mathbb{E}_1[\delta \mathbf{1}_{\{u \geq h\}}] \leq h^{-m} \mathbb{E}_1[\delta u^m] = O([\log(1/\varepsilon)]^{-m}).$$

Uniform boundedness makes the reward contribution on $u \geq h$ negligible after multiplication by $[\log(1/\varepsilon)]^{\rho_R}$. For each $\eta > 0$, the uniform local expansion gives, for sufficiently small h and $A - A_*$,

$$(a_0 - \eta)u^{\rho_R} \leq \mathcal{R}_A(u) \leq (a_0 + \eta)u^{\rho_R}, \quad 0 < u < h.$$

The upper residual-moment bound and the attaining conditional-error rule give matching upper and lower reward bounds. The discarded $u \geq h$ part of the ρ_R -moment is also negligible, since it is bounded by $h^{\rho_R - m} \mathbb{E}_1[\delta u^m]$. Letting $\eta \downarrow 0$ yields

$$\sup_{(S, \delta)} \mathbb{E}_1[\mathcal{R}_A(A - S)\delta] \sim \frac{a_0 \mathcal{M}_{\alpha, \beta}(\rho_R)}{[2 \log(1/\varepsilon)]^{\rho_R}}.$$

The value at $u = 0$ contributes nothing because $\mathcal{R}_A(0) = 0$. This proves (6.2).

E.5. Monitoring grids and implementation lead

The sharp law allows the preterminal monitoring date to approach the horizon with ε . A fixed finite grid with a final preterminal gap bounded away from zero defines a different local asymptotic regime. If every counted certificate must precede economic death by a fixed information-time lead $\ell > 0$, then the observation horizon available for certification is $A - \ell$ and the feasibility boundary shifts to $A_* + \ell$. Calendar-time lead and information-time lead coincide only under a unit-speed information clock.

E.6. Equal lifetime information, different cash capacity

On $[0, 1]$ define

$$\Delta^{(1)}(t) = 1 - t, \quad \Delta^{(2)}(t) = (1 - t)^2(1 + b_{\text{poly}}t + a_{\text{poly}}t^2),$$

where

$$a_{\text{poly}} := \frac{135 - 5\sqrt{345}}{8}, \quad b_{\text{poly}} := 2 - \frac{2a_{\text{poly}}}{5}.$$

Direct integration gives

$$\Delta^{(1)}(0) = \Delta^{(2)}(0) = 1, \quad \int_0^1 \Delta^{(1)}(t) dt = \int_0^1 \Delta^{(2)}(t) dt = \frac{1}{2},$$

and

$$\int_0^1 [\Delta^{(1)}(t)]^2 dt = \int_0^1 [\Delta^{(2)}(t)]^2 dt = \frac{1}{3}.$$

With the same constant volatility, the two opportunities have the same current edge, physical horizon, integrated future cash value, lifetime information and CR . To make the near-critical family explicit, take the common volatility $\sigma_A^2 = 1/(3A)$ and let $A \downarrow A_*$. Both experiments then have information horizon A , with positive limiting volatility and fixed terminal profiles satisfying the uniform reward hypotheses. Their local crossing orders are $p = 1$ and $p = 2$, so (6.4) gives

$$\text{CAC}_1^{\text{cash}}(A_* + \varepsilon) \asymp [\log(1/\varepsilon)]^{-2/3},$$

$$\text{CAC}_2^{\text{cash}}(A_* + \varepsilon) \asymp [\log(1/\varepsilon)]^{-3/5}.$$

Lifetime information determines feasibility; cash value also depends on the payoff map.

F. Proofs for unknown decay and model transfer

F.1. Ordered least-profile family

Let $\underline{\Delta}$ and $\underline{A}_t$ be as in Section 7.1. The standardized matched statistic

$$\bar{Z}_T^{\text{lf}} := \frac{\int_0^T \underline{\Delta}(u) dY_u}{\sigma^2 \sqrt{\underline{A}}}$$

is $N(0, 1)$ under the boundary law. Under alternative ϑ its mean is

$$\frac{\int_0^T \underline{\Delta}(u) \Delta_{\vartheta}(u) du}{\sigma^2 \sqrt{\underline{A}}} \geq \sqrt{\underline{A}}.$$

Therefore $\mathbf{1}_{\{\bar{Z}_T^{\text{lf}} \geq z_{1-\alpha}\}}$ has size α and worst-case power $\Phi(\sqrt{\underline{A}} - z_{1-\alpha})$, attained at the least member. No level- α rule can have larger worst-case power because the least member belongs to the family. For the sharp capacity statement, apply the conditional-error crossing policy from Appendix E on the deterministic least-profile clock. Coupling each alternative and the least member with the same Brownian path gives

$$\underline{Z}_t^{\vartheta} - \underline{Z}_t^{\text{lf}} = \frac{1}{\sigma^2} \int_0^t \underline{\Delta}(u) [\Delta_{\vartheta}(u) - \underline{\Delta}(u)] du \geq 0.$$

Hence every least-profile crossing occurs no later under the other alternatives. Residual least-profile information is then pathwise larger. The same is true of the coupled payoff for a common nonnegative nondecreasing residual reward, or for the memberwise-dominating payoff maps specified in the theorem. The least member supplies the matching upper bound, so the sharp constants coincide with the simple least-profile constants.

F.2. Least-favorable mixtures

For the finite Gaussian experiment of Section 7.2, define

$$\mathcal{D}_\alpha := \{\delta \in L^\infty(\mathbb{P}_0) : 0 \leq \delta \leq 1, \mathbb{E}_0 \delta \leq \alpha\}.$$

For $w \in \mathcal{W}_J$,

$$\mathbb{E}_0[L_w \delta] = \sum_{j=1}^J w_j \mathbb{E}_j \delta$$

is continuous and bilinear in (δ, w) . The decision set is weak-star compact and convex, and $\mathcal{W}_J$ is compact and convex. The minimax theorem yields

$$\sup_{\delta \in \mathcal{D}_\alpha} \min_j \mathbb{E}_j \delta = \min_{w \in \mathcal{W}_J} \sup_{\delta \in \mathcal{D}_\alpha} \mathbb{E}_0[L_w \delta].$$

The inner problem is a Neyman–Pearson test against the mixture likelihood L_w .

If two distinct alternatives satisfy $\|m_1\|^2 = \|m_2\|^2 = A_*$, each has individual maximal power $1 - \beta$. A common rule attaining that power under both would be a most-powerful size- α test for each member. Continuous Gaussian likelihoods make each most-powerful test unique up to null sets, whereas distinct equal-norm vectors define distinct rejection halfspaces. Hence the individual frontiers cannot be attained simultaneously.

F.3. Same-contract approximation

Let the policy notation be as in Section 7.3. Tightened approximate feasibility implies

$$\text{size}(\hat{\pi}) \leq \widehat{\text{size}}(\hat{\pi}) + \eta_P \leq \alpha,$$

$$\text{pow}_j(\hat{\pi}) \geq \widehat{\text{pow}}_j(\hat{\pi}) - \eta_P \geq 1 - \beta \quad \text{for every } j.$$

Define

$$\Pi(a, p) := \{\pi \in \Pi : \text{size}(\pi) \leq a, \inf_j \text{pow}_j(\pi) \geq p\}, \quad \mathcal{U}^*(a, p) := \sup_{\pi \in \Pi(a, p)} \mathcal{U}(\pi).$$

Assume $2\eta_P < \min(\alpha, \beta)$ and that $\Pi(\alpha - 2\eta_P, 1 - \beta + 2\eta_P)$ is nonempty. Suppose $\hat{\pi}$ is ε_{opt} -optimal for the tightened approximate problem with objective $\hat{\mathcal{U}}$. Uniform memberwise reward error implies $|\mathcal{U}(\pi) - \hat{\mathcal{U}}(\pi)| \leq \eta_V$. Every policy in the twice-tightened true class is feasible for the once-tightened approximate problem. Comparing $\hat{\pi}$ to an arbitrary such policy, and taking the supremum, gives

$$\mathcal{U}(\hat{\pi}) \geq \mathcal{U}^*(\alpha - 2\eta_P, 1 - \beta + 2\eta_P) - 2\eta_V - \varepsilon_{\text{opt}}.$$

Consequently,

$$\begin{aligned} \mathcal{U}^*(\alpha, 1 - \beta) - \mathcal{U}(\hat{\pi}) &\leq 2\eta_V + \varepsilon_{\text{opt}} \\ &\quad + \mathcal{U}^*(\alpha, 1 - \beta) - \mathcal{U}^*(\alpha - 2\eta_P, 1 - \beta + 2\eta_P). \end{aligned}$$

The last difference is the loss due to tightening the reliability contract. This argument guarantees the fixed worst-case criterion; each member's separate optimum remains outside this guarantee. A sufficient probability bound is obtained from total variation, with convention $\text{TV}(P, Q) := \sup_E |P(E) - Q(E)|$, where the supremum is over measurable events. If

$$\max_{j \in \{0, 1, \dots, J\}} \text{TV}(P_j, \hat{P}_j) \leq \eta_P,$$

then every bounded history-measurable deployment probability changes by at most η_P . If the reward range has length at most V_{span} and the approximate payoff differs pointwise by at most κ_V , one may take

$$\eta_V = V_{\text{span}}\eta_P + \kappa_V.$$

Pinsker's inequality gives

$$\text{TV}(P_j, \hat{P}_j) \leq \sqrt{\frac{1}{2} \text{D}_{\text{KL}}(P_j \| \hat{P}_j)}.$$

For finite-dimensional Gaussian history laws with common positive-definite covariance Σ and means $m, \hat{m}$,

$$\text{TV}\{N(m, \Sigma), N(\hat{m}, \Sigma)\} = 2\Phi\left(\frac{\|\Sigma^{-1/2}(m - \hat{m})\|}{2}\right) - 1.$$

These inequalities put model-estimation error and the certification margin on the same probability scale.

G. Notation and scope of the models

Decorations distinguish mathematical roles; local model assumptions remain part of each result.

Symbol	Meaning and scope
$\mathbb{P}_0, \mathbb{P}_1; U_{\text{rnd}}$	Null/alternative laws; independent auxiliary uniform randomizer available at time zero.
α, β	Size and nondeployment bounds; sharp asymptotics require $0 < \alpha < 1 - \beta < 1$.
$A_t; H = A_T; A$	Likelihood quadratic-variation clock; random terminal clock; deterministic horizon.
$t_{\text{inv}}(s); Z_s$	Stopped inverse clock; log likelihood in information time.
$S, \delta; \rho_{\text{dep}}$	Stopping time and binary decision; convexified deployment probability in $[0, 1]$.
$c_{\alpha, \beta}(A)$	Reliable residual-information capacity, reported as zero below feasibility.
$A_*, \mathcal{I}_*$	Critical information time and KL information, $A_* = 2\mathcal{I}_*$.
$\Delta; c, p$	Boundary drift gap; terminal coefficient (vector or scalar as stated) and crossing order $p > 0$.
$\lambda_{\text{dec}}; \lambda_{\text{Lag}}, \omega$	Decay rate; nonnegative size/power dual multipliers.
$\lambda_q; \lambda_{\text{tr}}$	Predictable-certificate tilt; quadratic turnover-cost coefficient.
$v_{\text{KL}}; v_R, v_X$	Value per KL nat; return and predictor innovation variances.
$\rho_{\text{corr}}; \rho_R$	Innovation correlation; positive residual-reward exponent.
$\kappa_{\text{crowd}}; \kappa_{\text{KL}}$	Linear crowding sensitivity; stationary pairwise information rate.
$L_{\text{hor}}; L_{\text{score}}$	Discrete horizon; empirical lifetime score with descriptive scope.
L_j, L_w	Alternative and mixture likelihood ratios in the finite robust experiment.
$\mathcal{U}(\pi); \eta_P, \eta_V$	Worst-member transfer value; uniform probability/reward approximation bounds.
$\mathcal{K}_{\alpha, \beta}; \mathcal{M}_{\alpha, \beta}(\rho_R)$	Sharp inverse-logarithmic constant; critical Gaussian positive-part moment.
$\mathcal{R}_A(u); K_p^{\text{unit}}$	Residual reward with the stated uniform local expansion; unit-exposure reward coefficient.